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Final Project Report

MEE7012-SUM

3D Printing Parameter Optimisation: Impact of K-Value on Carbon-Reinforced PETG

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Summary

This project investigated the influence of flow-rate multiplier (K) and build orientation on the tensile behaviour of PETG and carbon-fibre-reinforced PETG (CF-PETG) produced using fused deposition modelling (FDM). The engineering relevance of the study lies in establishing how a key material-flow parameter affects the mechanical performance of additively manufactured components and identifying suitable processing conditions for different materials.

Specimens were manufactured using four K-values (80%, 90%, 100% and 110%) in both flat and on-edge orientations. Tensile testing was conducted using ISO 527-5A specimens, with stress calculated using the measured specimen cross-sectional area. Young's modulus, ultimate tensile strength (UTS), elongation at break and toughness were evaluated. Representative stress–strain curves were selected using a composite-score approach, while optical microscopy was used to examine the printed and fractured material and identify visible structural defects. Statistical analysis was used to determine the significance of material, K-value and orientation effects and their interactions.

The results showed that CF-PETG was substantially stiffer than PETG and exhibited a stronger response to changes in K, particularly in UTS. For CF-PETG, K = 110% produced a significantly higher UTS than K = 80%, 90% and 100%, while Young's modulus varied comparatively little across the investigated K-values. PETG showed comparatively stable UTS and modulus but substantially greater specimen-to-specimen variation in elongation and toughness, associated with the presence of both limited/brittle and extensive-drawing behaviour. Build orientation produced some differences in the observed responses, although its effect was dependent on the material and property considered.

Overall, the findings indicate that increasing K can improve tensile strength in CF-PETG within the investigated range, but increasing flow does not provide a universal improvement across all materials and mechanical properties. Therefore, K-value selection should be based on the material and the specific mechanical performance required.

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Nomenclature

Symbols:

A	Cross-sectional area of the specimen gauge section (m^2)
E	Young's modulus (Pa)
F	Applied tensile force (N)
K	Flow-rate multiplier (%)
L_0	Initial gauge length (m)
L_f	Gauge length at specimen failure (m)
t	Specimen thickness (m)
w	Specimen width (m)
σ	Engineering stress (Pa)
ε	Engineering strain
ε_b	Elongation at break (%)
$\Delta\sigma$	Change in engineering stress (Pa)
$\Delta\varepsilon$	Change in engineering strain
UT	Toughness (J/m^3)

Subscripts:

0	Initial/reference condition
b	At break
f	At failure
i	Data-point index
n	Total number of data points

Abbreviations:

AM	Additive manufacturing
ANOVA	Analysis of variance
CF-PETG	Carbon-fibre-reinforced polyethylene terephthalate glycol
FDM	Fused deposition modelling
HSD	Honestly significant difference
IQR	Interquartile range
ISO	International Organization for Standardization
PET	Polyethylene terephthalate
PETG	Polyethylene terephthalate glycol

SEM	Scanning electron microscopy
STL	Standard Tessellation Language
UFP	Ultimaker Format Package
UTS	Ultimate tensile strength
p	Statistical significance probability
R ²	Coefficient of determination

1.0 Introduction

1.1 Background to Additive Manufacturing

Additive manufacturing (AM) is a manufacturing approach in which three-dimensional components are produced through the successive addition of material based on digital three-dimensional model data. This distinguishes AM from conventional subtractive and formative manufacturing processes, where material is removed or shaped from an existing workpiece (ISO/ASTM, 2021). The development of AM is closely associated with rapid prototyping (RP), which initially enabled physical prototypes to be produced directly from digital designs without the need for conventional tooling. This allowed designers to evaluate and modify products more efficiently during the development process (Gibson et al., 2015).

Since its early use for rapid prototyping, AM has developed towards the production of functional and end-use components. Improvements in materials, machine capabilities and digital design have expanded its application across sectors including aerospace, automotive, medical and consumer products (Tofail et al., 2018). AM is particularly suited to customised and geometrically complex components that may be difficult or inefficient to manufacture using conventional processes (Tofail et al., 2018, Ngo et al., 2018).

A major advantage of AM is its ability to produce complex geometries directly from digital models while reducing the need for dedicated tooling and potentially reducing material waste (Ngo et al., 2018, Tofail et al., 2018). However, the quality and mechanical performance of AM components depend on the manufacturing process and its processing conditions. Challenges associated with AM include internal void formation and anisotropic behaviour, while dimensional accuracy and surface characteristics are also important considerations in achieving functional components (Ngo et al., 2018, Tofail et al., 2018).

AM encompasses several process families, including vat photopolymerization, material extrusion, material jetting, powder bed fusion, binder jetting, directed energy deposition and sheet lamination (ISO/ASTM 52900:2021). These processes differ in their feedstock materials and methods of material deposition. The main process families of additive manufacturing are illustrated in Figure 1.1 (Dhanunjayarao et al., 2020). The present study focuses on material extrusion, with the specific FDM process and its associated processing parameters discussed in Section 2.1.

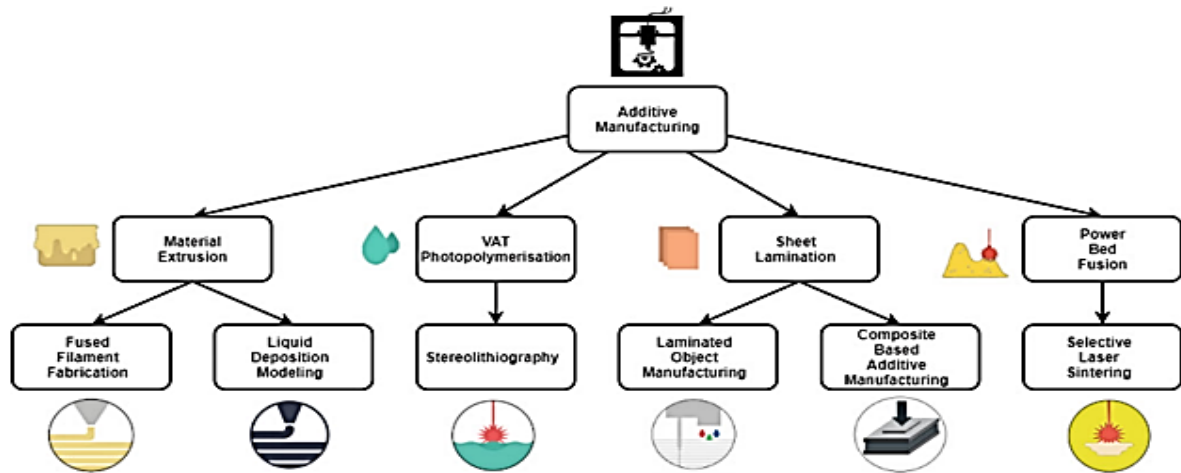


Figure 1.1 Classification of different Additive Manufacturing Technologies. Reproduced from Dhanunjayarao et al. (2020)

1.2 PETG and CF-PETG in Engineering Applications

Polyethylene terephthalate glycol (PETG) is a glycol-modified form of polyethylene terephthalate (PET) and is widely used in both conventional polymer products and additive manufacturing. PETG combines good toughness, chemical resistance and processability, making it suitable for a variety of everyday and engineering applications. In conventional applications, PETG is used for products such as bottles, containers, packaging and transparent components, while in additive manufacturing it is commonly used to produce housings, brackets, fixtures, covers and customised components.

The suitability of PETG for additive manufacturing also makes it useful where relatively complex geometries or customised designs are required. Figure 1.2(a) shows examples of PETG components produced using additive manufacturing, illustrating the range of geometries that can be produced from the material. Its combination of mechanical performance and printability has contributed to its use in functional rather than purely visual prototypes (Cano-Vicent et al., 2021).

Carbon-fibre reinforcement provides a means of modifying the characteristics of PETG for applications where greater stiffness and dimensional stability are required. CF-PETG is produced by incorporating short carbon fibres into the polymer matrix, resulting in a composite material with different mechanical behaviour from unreinforced PETG. Such materials have attracted interest in lightweight brackets, fixtures, mechanical components and other applications where increased rigidity is beneficial (Kichloo et al., 2021, Mangalampalli et al., 2025). An example of a 3D-printed mechanical component is shown in Figure 1.2(b).

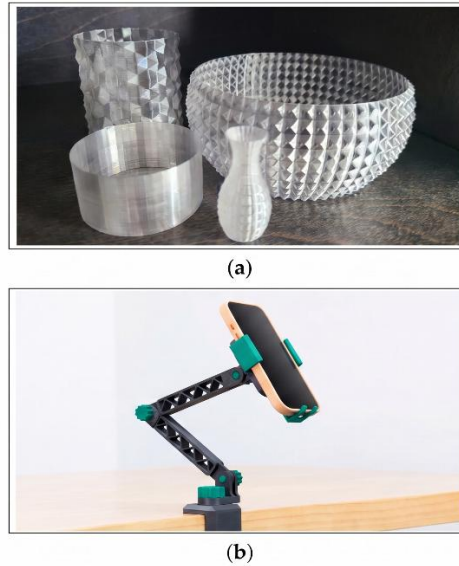


Figure 1.2 Examples of 3D-printed components manufactured from (a) PETG and (b) CF-PETG.

The use of PETG and CF-PETG in this study therefore provides a comparison between an unreinforced thermoplastic and a carbon-fibre-reinforced polymer. This comparison is particularly relevant when considering how changes in printing conditions may affect materials with different reinforcement structures. The detailed material characteristics, mechanical properties and reported FDM performance of PETG and CF-PETG are discussed separately in Section 2.2.

1.3 Flow-Rate Multiplier (K Value) in FDM

During FDM printing, the slicing software determines the nominal amount of material to be extruded along each toolpath based on the part geometry and selected printing parameters. The flow-rate multiplier, K , scales this nominal extrusion amount and, in this study, is expressed as a percentage of the nominal flow. A value of $K = 100\%$ represents the nominal flow, while values below or above 100% reduce or increase the specified material flow, respectively.

Changes in the K therefore alter the amount of material deposited during printing. A reduced flow rate can result in insufficient material deposition, while an increased flow rate can lead to excessive material deposition. The general relationship between reduced, desired and increased extrusion is illustrated in Figure 1.3. (Huang, 2018).

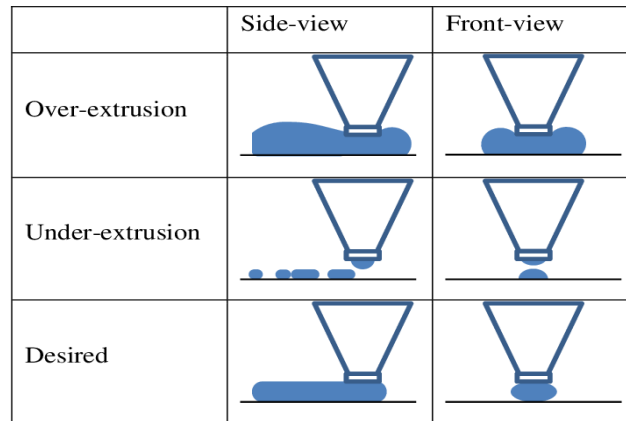


Figure 1.3 Schematic representation of under-extrusion, desired extrusion and over-extrusion during material extrusion. Reproduced from Huang (2018)

In the present study, the K-value was investigated as the primary process variable at four levels: $K = 80\%$, $K = 90\%$, $K = 100\%$, and $K = 110\%$. These values were used to investigate the influence of reduced, nominal and increased material flow on the tensile properties of PETG and CF-PETG across the selected build orientations.

1.4 Aim and Objectives

Aim

This study aims to determine how the K and build orientation influence the tensile properties of FDM-printed PETG and carbon-fibre-reinforced PETG (CF-PETG), and whether the response to these two process variables differs between an unreinforced polymer and a fibre-reinforced composite.

The objectives of this project are:

- Review the literature on flow rate and build orientation in FDM and identify the gap regarding their combined effect on an unreinforced and a fibre-reinforced material.
- Manufacture PETG and CF-PETG tensile specimens across four K conditions and two build orientations.
- Determine the tensile properties of each specimen condition, including ultimate tensile strength, Young's modulus, elongation at break and toughness.
- Examine the surface and fracture morphology of tested specimens using optical microscopy.
- Evaluate the combined effect of flow rate and build orientation on the tensile performance of PETG and CF-PETG.
- Compare the findings with the published literature and consider their implications for material efficiency and sustainable FDM processing.
- To consider the implications of the findings for material efficiency and sustainable FDM processing, and to identify opportunities for future work.

2.0 Technology Review/Literature Review

2.1 Fused Deposition Modelling (FDM) – Process Overview

Fused deposition modelling (FDM) is a material extrusion additive manufacturing process in which thermoplastic material is deposited through a heated nozzle to construct a component layer by layer. A thermoplastic filament is fed into the print head, where it is heated and subsequently extruded through the nozzle onto the build platform. The print head moves according to a programmed toolpath, while relative movement between the print head and build platform allows successive layers to be deposited to form the final component, as illustrated in Figure 2.1 (Singh et al., 2023).

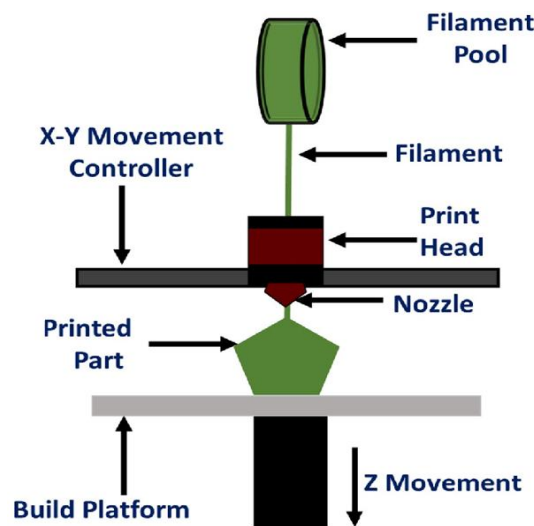


Figure 2.1 Schematic representation of the fused deposition modelling (FDM) process.

Reproduced from Singh et al. (2023)

The deposited material forms individual extrusion beads that bond with previously deposited material during the printing process. The quality of this bonding is influenced by the thermal history of the deposited polymer. At the interface between adjacent deposited regions, polymer chains can diffuse across the interface and become entangled, allowing a bonded structure to develop (Bellehumeur et al., 2004). Sufficient thermal energy and time are therefore required for effective interfacial diffusion and coalescence. Rapid cooling of the deposited material can restrict this process and result in weaker interfaces between deposited layers (Bellehumeur et al., 2004, Omer et al., 2025).

FDM is widely used because of its relatively simple processing requirements, comparatively low equipment cost and compatibility with a broad range of thermoplastic materials (Cano-Vicent et al., 2021). These characteristics have contributed to the widespread use of FDM for both rapid

prototyping and the production of functional components in sectors such as automotive, aerospace and consumer products.

However, the layer-by-layer nature of FDM means that the resulting component does not necessarily exhibit the same mechanical behaviour in all directions. The strength and stiffness of a printed component can depend on the direction of loading relative to the deposited material and the interfaces between layers. Consequently, the mechanical performance of FDM-printed components is influenced by both build orientation and processing conditions. These factors are considered in the following sections, with particular attention given to the material systems investigated in this study and the influence of build orientation and flow rate.

2.2 PETG and CF-PETG: Properties and FDM Performance

Polyethylene terephthalate glycol (PETG) is a glycol-modified copolyester widely used in FDM printing because of its combination of strength, ductility and processability. Compared with other commonly used FDM thermoplastics such as PLA and ABS, PETG offers a useful balance of mechanical performance and printability, making it suitable for both prototyping and functional components (Kichloo et al., 2021, Cano-Vicent et al., 2021). Its relatively good interlayer adhesion and low shrinkage during printing also contribute to its suitability for additive manufacturing applications (Cano-Vicent et al., 2021).

Carbon-fibre reinforcement can be used to modify the mechanical behaviour of PETG. CF-PETG is commonly produced using short, chopped carbon fibres dispersed within the PETG polymer matrix. The addition of carbon fibres generally increases the stiffness of the material and can also influence its tensile strength, while reducing its ability to undergo deformation before fracture. Kichloo et al. (2021) reported a substantial increase in elastic modulus following carbon-fibre reinforcement of PETG, while other studies have similarly reported increases in tensile strength and modulus accompanied by reduced elongation at break (Alarifi, 2023, Balasrinivasan et al., 2024). These changes reflect the reinforcing effect of the carbon fibres and the resulting difference in mechanical behaviour between the unreinforced polymer and the composite.

The differences in fracture morphology between PETG and CF-PETG can also be observed through scanning electron microscopy (SEM). Prabhakaran et al. (2025) reported a characteristic river-like fracture pattern for pure PETG, as shown in Figure 2.2(a). In comparison, the 20% CF-PETG specimen exhibited voids within the polymer matrix, as shown in Figure 2.2(b). The authors also reported

differences in layer formation, layer alignment and bonding between the materials, demonstrating the influence of the composite structure on the resulting fracture morphology (Prabhakaran et al., 2025).

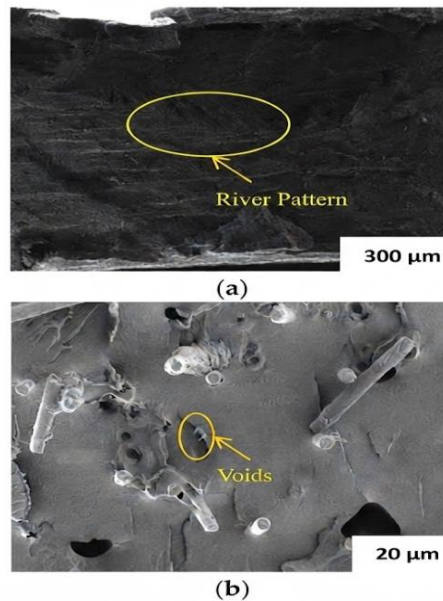


Figure 2.2 SEM images of (a) pure PETG showing a characteristic river pattern and (b) 20% CF-PETG showing voids. Reproduced from Prabhakaran et al. (2025)

The mechanical properties of PETG and CF-PETG reported in the literature vary considerably between studies. This variation can result from differences in material grade, printing parameters, specimen geometry, build orientation and tensile testing conditions. The testing standard used can also influence the reported values, as studies may employ different standards such as ASTM D638 and ISO 527-2, which specify different specimen geometries and testing requirements (Szykiedans et al., 2017). Consequently, reported mechanical properties should be considered indicative of the behaviour of the material and processing conditions investigated rather than as directly comparable absolute values.

For FDM applications, the presence of carbon fibres also introduces additional processing considerations. Carbon-fibre-reinforced filaments are more abrasive than unreinforced thermoplastics and therefore require suitable nozzle materials to reduce nozzle wear. The fibres can also influence the flow and deposition behaviour of the molten composite, while their orientation within the printed structure can contribute to anisotropic mechanical behaviour. Consequently, the mechanical performance of CF-PETG depends not only on the inherent reinforcement provided by the carbon fibres but also on the processing conditions used during FDM.

Overall, PETG and CF-PETG provide two contrasting material systems for investigating the influence of FDM processing conditions. PETG represents an unreinforced thermoplastic with relatively high ductility, whereas CF-PETG provides a stiffer composite system in which carbon-fibre reinforcement can alter both the mechanical response and fracture morphology. This distinction provides a basis for comparing the influence of the K and build orientation on the tensile behaviour of the two materials in the present study. The specific PETG and CF-PETG materials used in this investigation, together with their manufacturer-reported properties, are presented in Section 3.1.

2.3 Process Parameters Affecting Mechanical Performance

The mechanical performance of FDM-printed parts is influenced by a range of process parameters that govern how material is deposited and how successive layers bond to one another. This section briefly reviews the parameters most commonly investigated in the literature, before flow rate, the primary variable of this study, is examined in more detail in Section 2.4. The principal process parameters considered in FDM, including build orientation, raster direction and layer height, are illustrated schematically in Figure 2.3.

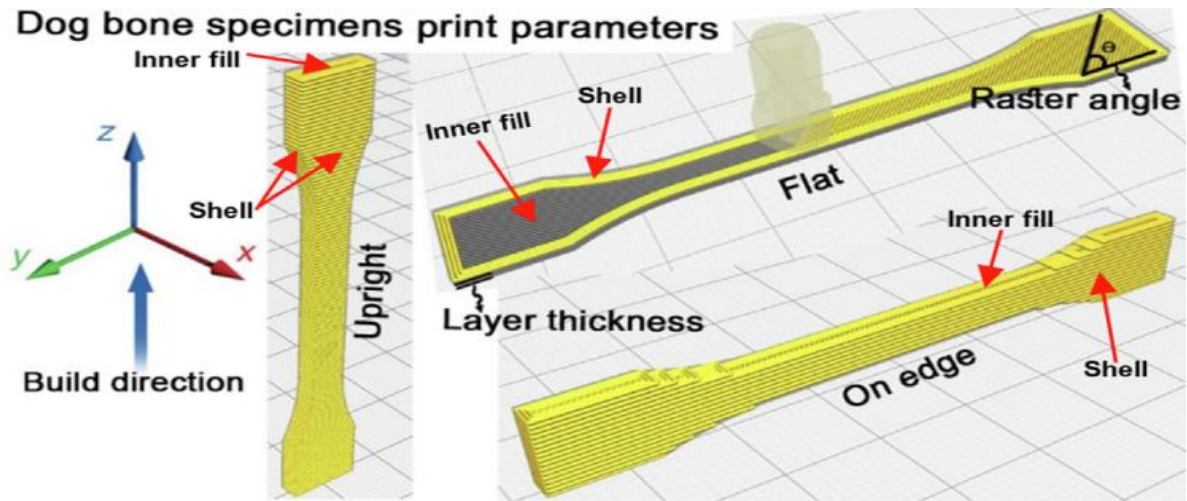


Figure 2.3 Schematic representation of printing parameters for FDM-printed dog-bone specimens.

Reproduced from Hanon et al. (2021)

Layer height determines the number of individual layers, and therefore the number of interlayer interfaces, within a printed part. Stojkovic et al. (2023) found that layer height had a significantly greater effect on tensile strength than either annealing time or temperature when comparing PLA, PETG, and CF-PETG specimens, with both PETG and CF-PETG achieving optimal tensile strength at the smallest layer height tested (0.1 mm). Patel et al. (2024) investigated layer thicknesses of 0.2, 0.3 and 0.4 mm for carbon-fibre-reinforced PETG and found that 0.2 mm formed part of the parameter combination producing the highest mean flexural performance within the tested range.

Print speed affects the thermal history of the deposited material by changing the time available for heat transfer and cooling during deposition. Since interlayer bonding depends on the thermal history of the polymer interface, changes in print speed can influence the degree of bonding between adjacent layers Bellehumeur et al. (2004). Patel et al. (2024) investigated printing speeds of 40, 50 and 60 mm/s for carbon-fibre-reinforced PETG and identified printing speed as an important parameter affecting mechanical performance. Within their tested parameter range, 40 mm/s formed part of the parameter combination producing the highest mean flexural performance.

Infill density and pattern determine the internal structure of a printed part away from its outer walls. Rihar and Hozdic (2026), using the same ISO 527-2 standard and 5 mm/min test speed adopted in this study, tested PETG and PETG-CF specimens and found that increasing linear infill density from 30% to 100% increased PETG tensile strength from 16.99 ± 0.52 MPa to 31.35 ± 0.33 MPa. Hozdić (2024) similarly investigated PETG and PETG-CF at 30%, 60% and 100% infill densities and reported that infill density significantly affected the measured mechanical response of both materials. The latter study reported changes in maximum force rather than tensile strength and is therefore used here only as evidence that infill density affects mechanical behaviour rather than as a quantitative tensile-strength comparison.

Nozzle diameter influences mechanical performance by controlling the width of each deposited bead and the degree of interlayer contact and fusion. Hodzic and Pandzic (2019) et al. (2024) reported that increasing nozzle diameter from 0.4 to 0.6 mm affected the mechanical performance of FDM-printed PET-CF, increasing tensile strength and elastic modulus by up to 21.6% and 17.6%, respectively, while reducing strain at break, reflecting the trade-off between stiffness and ductility associated with larger nozzle diameters in carbon-fibre-filled materials.

Build orientation determines the relationship between the deposited layers, extrusion paths and the direction of applied loading. Hanon et al. (2021) compared flat, on-edge and upright orientations using FDM-printed dog-bone specimens and demonstrated that build orientation can influence the dimensional characteristics of printed specimens. Faidallah et al. (2023) investigated the tensile behaviour of PETG specimens manufactured in flat and on-edge orientations; for the ISO 527-2 specimens, tensile strength increased from 13.4 MPa for the flat orientation to 17.1 MPa for the on-edge orientation, corresponding to an increase of approximately 27.6%. This flat-versus-on-edge comparison is particularly relevant to the present study because the same two build orientations are investigated using ISO 527-2 Type 5A specimens.

Raster direction influences the orientation of deposited material within each layer and can therefore affect the direction in which load is transferred through the printed structure. Patel et al. (2024) investigated raster angles of 0°, 45° and 90° and identified raster angle as an important parameter affecting the mechanical performance of carbon-fibre-reinforced PETG. Elayeb et al. (2023) specifically investigated raster angle in short carbon-fibre-reinforced PET-G and reported an average ultimate tensile strength of 52.2 MPa at 0° compared with 25.4 MPa at 90°, demonstrating the influence of raster orientation on tensile behaviour. However, the effect of raster direction is not necessarily independent of other printing conditions. Hanon et al. (2021) found that raster direction had little influence on the dimensional accuracy of their dog-bone specimens when the number of layers and contour size were maintained.

Overall, the literature demonstrates that FDM mechanical performance is affected by the interaction of several process parameters rather than by a single printing condition. Layer height, printing speed, infill, nozzle diameter, raster direction and build orientation can each modify the structure and thermal history of a printed component. For the present study, these parameters were therefore maintained at fixed values so that the K multiplier could be investigated as the principal experimental variable.

2.4 Research on Flow Rate in FDM

The flow-rate multiplier (K) scales the volume of material extruded relative to the nominal amount calculated by the slicer for a given toolpath. It is therefore one of the more direct process controls affecting the amount of material deposited within an FDM-printed component. Changes in flow rate can influence bead geometry, the contact between adjacent extrusion paths and the formation of internal voids and consequently may affect the mechanical performance of printed components. The influence of flow rate has been investigated across different thermoplastic and fibre-reinforced material systems using a range of extrusion-multiplier values.

Barrios and Romero (2019) examined flow rates between 90% and 110% for PETG and found that flow rate affected surface roughness and geometric accuracy, with deviations from 100% influencing the correspondence between the deposited bead width and the nominal toolpath width. Lendvai et al. (2024) investigated extrusion multipliers from $K = 0.97$ to 1.05 for PLA and found that increasing the extrusion multiplier reduced internal porosity from approximately 5.82% to 0.05%. The number of detected defects also decreased as the extrusion multiplier increased, as

illustrated in Figure 2.4. Their results therefore demonstrate a clear relationship between the amount of material deposited and the internal structure of the printed specimens.

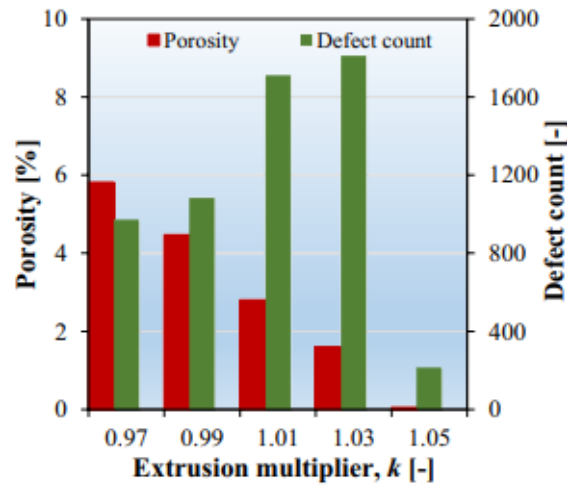


Figure 2.4 Porosity and detected defect count of specimens produced at different extrusion multiplier values. Reproduced from Lendvai et al. (2024)

Lendvai et al. (2024) also reported that tensile, flexural and impact strength improved with increasing extrusion multiplier up to $K = 1.03$, before decreasing at $K = 1.05$. This indicates that increasing material deposition does not necessarily produce a continuous improvement in mechanical performance, with an optimum occurring within the investigated range. Gao et al. (2026) similarly investigated flow rate alongside layer thickness, nozzle temperature and print speed across 256 parameter combinations for carbon-fibre-reinforced PLA (PLA-CF), using a machine-learning-based model to relate the combined parameters to tensile performance. Their results further identified flow rate as one of the parameters having a systematic influence on mechanical properties.

For materials more directly comparable to the present study, Minaoglou et al. (2024) investigated flow rate at 85%, 100% and 115%, alongside wall thickness and layer height, for CARBON:PLUS, a 20% carbon-fibre-reinforced PETG filament. A full factorial design containing 27 parameter combinations was used, allowing the influence of flow rate to be examined together with the other varied parameters. The study reported that flow rate contributed to changes in tensile performance.

The relationship between flow rate and tensile stress is illustrated by the three-dimensional response surfaces presented by Minaoglou et al. (2024) in Figure 2.5. The plots show the relationship between flow rate and tensile stress in combination with layer thickness and wall

thickness. These results are particularly relevant to the present investigation because they demonstrate the influence of flow rate on the mechanical response of a carbon-fibre-reinforced PETG material. However, because flow rate was varied alongside other process parameters, the individual effect of flow rate could not be considered completely independently.

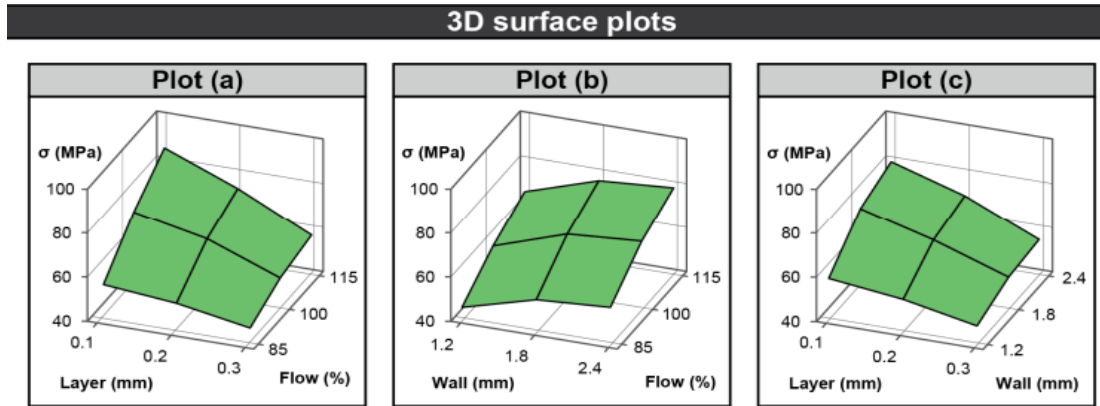


Figure 2.5 3-D surface plots showing the relationship between flow rate and other printing parameters and the resulting tensile stress of carbon-reinforced PETG. Reproduced from Minaoglou et al. (2024).

Economides et al. (2024) also investigated the influence of printing parameters on the mechanical properties of CF-PETG (Carbon-P), including factors associated with porosity and infill. Their results indicated that internal void content was closely related to the tensile performance of the printed material. This is consistent with the mechanism through which changes in material flow can alter the internal structure of FDM-printed components.

Across the reviewed studies, a broadly consistent relationship can be identified between material flow, internal structure and mechanical performance. Under-extrusion, resulting from insufficient material deposition, can produce gaps between adjacent extrusion lines and increase internal porosity. Increasing the flow rate can reduce these voids and improve the continuity of deposited material, potentially improving mechanical performance. However, excessive material deposition can eventually produce dimensional inaccuracies, surface irregularities and other defects, meaning that increasing flow beyond an appropriate level does not necessarily continue to improve part performance.

The practical effect of varying the extrusion multiplier over a wider range is illustrated by Butt et al. (2022), who investigated material extrusion rates from 70% to 150% for graphene-enhanced PLA. The specimens produced at increasing extrusion rates showed progressively greater material deposition and increasingly pronounced bead formation, as shown in Figure 2.6. The study

therefore, demonstrates the substantial changes in deposited material that can occur when the extrusion multiplier is varied over a wide range.

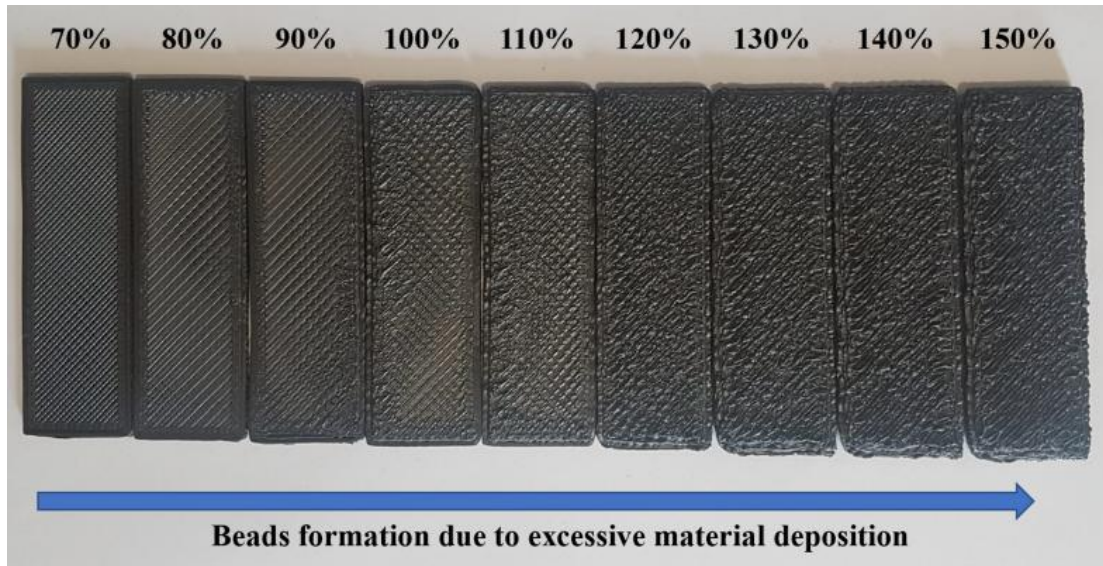


Figure 2.6 Printed graphene-enhanced PLA specimens at different flow rates (70–150%).

Reproduced from Butt et al. (2022).

Although the results of Butt et al. (2022) demonstrate the effect of a wide extrusion-multiplier range, the material system differs from the PETG and CF-PETG considered in this study. The findings should therefore be regarded as supporting evidence for the effect of extrusion variation rather than as directly transferable mechanical-performance data. More broadly, the reviewed literature indicates that material formulation can influence the response to changes in extrusion conditions, meaning that the effect of K variation may differ between unreinforced PETG and carbon-fibre-reinforced PETG.

The reviewed literature includes both relatively narrow ranges around the nominal flow rate and substantially wider ranges. Barrios and Romero (2019) investigated flow rates of approximately 90–110% for PETG, while Lendvai et al. (2024) investigated $K = 0.97$ – 1.05 for PLA. Minaoglou et al. (2024) investigated 85–115% flow for a 20% carbon-fibre-reinforced PETG filament. Butt et al. (2022) investigated a considerably wider range of 70–150% using graphene-enhanced PLA.

2.5 Research Gap

The reviewed literature demonstrates that flow rate can influence material deposition, internal porosity and the resulting mechanical performance of FDM-printed polymers and composites. However, the available studies differ substantially in terms of material system, flow rate range and combinations of other process parameters, limiting direct comparison between reported findings.

In particular, the effect of systematically varying the K-Value on both unreinforced PETG and carbon-fibre-reinforced PETG under otherwise consistent printing conditions remains insufficiently characterised. The interaction between flow rate and build orientation is also not well established for these two material systems. Consequently, it remains unclear whether changes in flow rate produce similar mechanical responses in PETG and carbon-fibre-reinforced PETG, or whether build orientation modifies these responses differently.

The present study addresses these gaps by comparing PETG and carbon-fibre-reinforced PETG across four K conditions, $K = 80\%$, 90% , 100% and 110% , and two build orientations, flat and on-edge. This produces a $2 \times 4 \times 2$ experimental matrices comprising 16 conditions, allowing the effects of K and build orientation to be evaluated for each material and enabling direct comparison between the unreinforced and fibre-reinforced systems.

3.0 Methodology

3.1 Filament Materials

Two filaments were used in this study: RS PRO PETG (RS Stock No. 1901962) and RS PRO CARBON-P (RS Stock No. 1901953), a 20% carbon-fibre-reinforced PETG. Both materials were purchased as 1 kg spools with a filament diameter of 2.85 mm. The PETG filament was supplied in white, while the CARBON-P filament was supplied in black. A single spool of each material was used throughout the experimental programme, eliminating batch-to-batch filament variation as a potential source of variability. PETG was used as the baseline material, with CARBON-P selected for comparison, as discussed in Section 2.2.

The manufacturer-reported properties of the selected filaments are summarised in Table 3.1, based on the RS PRO technical datasheets (PRO, n.d.–b, PRO, n.d.–a). The reported tensile strength at yield was 50 MPa for PETG and 52.5 MPa for CARBON-P. The reported tensile modulus was 2020 MPa for PETG and 3800 MPa for CARBON-P, while the strain at break was reported as 23% and 8%, respectively. These values provide a reference for comparison with the experimentally determined tensile properties obtained in this study.

Table 3.1 Material properties of the selected filaments (PRO, n.d.–b, PRO, n.d.–a)

Property	PETG	CF-PETG
Colour	White	Black

Filament Diameter	2.85 mm	2.85 mm
Diameter Tolerance	±0.10 mm	±0.10 mm
Tensile Strength at Yield	50 MPa	52.5 MPa
Tensile Modulus	2020 MPa	3800 MPa
Yield Strain	6%	4.2%
Strain at Break	23%	8%
Specific Gravity	1.27 g/cm ³	1.19 g/cm ³
Recommended Nozzle Type	0.4 mm	0.6 mm

3.2 Specimen Design

The tensile test specimens were designed in accordance with ISO 527-2:2025 using the Type 5A geometry shown in Figure 3.1(a)(BSI, 2025). This specimen type was selected to allow comparison with other 3D-printed specimens reported in the literature, as discussed in Section 2.2. Its compact size also allowed multiple specimens to be printed on the Ultimaker S5 build plate within a single job, helping to maintain consistent printing conditions across each batch. A single specimen geometry was used throughout the experimental programme for all materials and K conditions investigated.

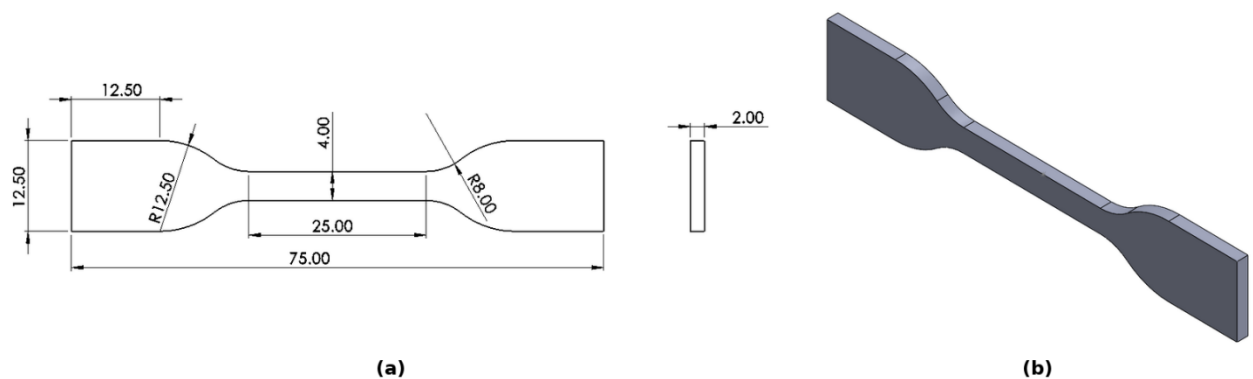


Figure 3.1 ISO 527-2 Type 5A specimen geometry: (a) engineering drawing and (b) SolidWorks isometric model.

The specimen was modelled in SolidWorks to the dimensions specified by the standard and is shown in Figure 3.1(b). Upon completion of the CAD model, the design was exported as an STL file. This format was selected as it is widely used for transferring CAD models to additive manufacturing software.

3.3 Specimen Manufacturing

3.3.1 Equipment

Two Ultimaker S5 fused deposition modelling (FDM) printers, located within the Student Design Centre at Queen's University Belfast, were used to print the specimens. One printer was dedicated to PETG and the other to CF-PETG, each fitted with the nozzle size appropriate to the material. A 0.4 mm nozzle was used for the PETG specimens, while a 0.6 mm nozzle was used for the CF-PETG specimens. The larger nozzle was selected to reduce wear caused by the abrasive carbon fibre reinforcement present in the CF-PETG filament. One of the two printers is shown in Figure 3.2.

Slicing and print preparation were carried out using Ultimaker Cura, version 5.12.0. The STL file described in Section 3.2 was imported into Cura, sliced according to the parameters given in Table 3.2, and exported as a .ufp (Ultimaker Format Package) file. This file bundles the generated G-code, the machine-readable instructions that direct the printer's movement, temperature, and extrusion, together with the associated print and material settings. The .ufp file was transferred to each printer via USB.



Figure 3.2 Ultimaker S5 printer used for specimen printing.

3.3.2 Printing Parameters

Print profiles were based on Cura's built-in Ultimaker PETG and Ultimaker PETCF material presets. A 0.4 mm nozzle at 245°C was used for PETG, while a 0.6 mm nozzle at 280°C was used for CF-PETG, with these values taken directly from the respective material presets. The bed temperature was set to 80°C and the fan speed to 10% for both materials. A layer height of 0.2 mm and a print speed of 40 mm/s were selected for both materials, as these values formed part of the parameter combination associated with the highest mean flexural performance reported by Patel et al. (2024) within their tested range for carbon-fibre-reinforced PETG. An infill density of 100% was used for both materials to maximise the nominal load-bearing cross-section and to remove infill density and

pattern as additional experimental variables. Wall configuration and infill pattern were kept consistent within each material, while K was varied between experimental conditions. The remaining printing parameters are summarised in Table 3.2.

Wall thickness was fixed at a nominal value of 2.4 mm for both materials. In Cura, the wall structure is defined using the wall line count and nozzle diameter. Therefore, six wall lines were used for PETG with the 0.4 mm nozzle and four wall lines for CF-PETG with the 0.6 mm nozzle, corresponding to nominal wall thicknesses of 2.4 mm in both cases. This provided the same nominal wall thickness for both materials despite the different nozzle diameters.

Because PETG and CF-PETG were printed using different nozzle diameters, nozzle temperatures and wall line counts, the absolute amount of material deposited at the same nominal K-value may differ between the two materials. Therefore, the effect of K was primarily assessed by comparing changes in tensile properties across K-values within each material, while differences between the two materials were considered in relation to their different printing conditions.

Table 3.2 Fixed printing parameters

Parameter	Values
Layer height (mm)	0.2 mm
Infill density (%)	100%
Print speed (mm/s)	40 mm/s
Infill Pattern	Lines
Wall thickness (mm)	2.4 mm

Both filaments were dried before use using a filament dryer set to 45°C for 6–7 hours, typically on the evening before a print session so that the filament was ready for use the following day. Before each print, the build plate was cleaned using water and a tissue, followed by application of a layer of Dimafix adhesive to improve bed adhesion.

3.3.3 Experimental Variables

This study investigated the effects of material, K-value and build orientation on the tensile properties of the printed specimens. K was adjusted through the Cura flow setting, with four levels investigated: K = 80%, 90%, 100% and 110%. The selected range was informed by previous studies

investigating flow-rate and extrusion-multiplier effects in PETG and carbon-fibre-reinforced PETG-based filaments (Barrios and Romero, 2019, Minaoglou et al., 2024).

Build orientation was investigated using two configurations, as shown in Figure 3.3. In the flat orientation (Figure 3.3(a)), the specimen's largest face was in contact with the build plate, whereas in the on-edge orientation (Figure 3.3(b)), the specimen was printed on its narrow edge. This orientation comparison was selected based on previous work showing a difference in tensile behaviour between flat and on-edge PETG specimens tested according to ISO 527-2 (Faidallah et al., 2023). The on-edge specimens required support material to maintain print stability during manufacturing.

Specimens were printed using Cura's "Print One at a Time" mode to prevent nozzle travel between specimens. This required the primary flat-orientation batches to be arranged with one specimen in a different plate position from the remaining specimens. Inspection of the sliced toolpath preview indicated that this rotation did not alter the toolpath within the gauge section, as the gauge width was fully occupied by wall lines for both nozzle sizes. To maintain consistent placement conditions across the batch, the specimen printed in this position was excluded from the test set. This issue did not affect the on-edge batches, which could be arranged without a comparable placement constraint.



Figure 3.3 Specimen batches printed in (a) flat orientation and (b) on-edge orientation.

3.3.4 Experimental Variables

A full factorial design of experiments was adopted for this study, with every combination of the three experimental factors tested. The experimental factors were material (PETG and CF-PETG), K-

value (80%, 90%, 100% and 110%), and build orientation (flat and on-edge). This resulted in 16 experimental conditions ($2 \times 4 \times 2 = 16$), allowing the individual and combined effects of the three factors, including their interactions, to be evaluated.

All remaining printing parameters were held constant across the experimental conditions, as summarised in Table 3.2. Nozzle size and nozzle temperature were fixed for each material but differed between PETG and CF-PETG; consequently, comparisons between materials cannot fully separate the effect of material composition from the effect of these material-specific parameter differences. The full factorial design allowed the effects of material, K-value and build orientation to be evaluated systematically using the tensile results obtained from each experimental condition.

3.4 Tensile Testing

3.4.1 Equipment

Tensile testing was carried out using a Lloyd Instruments LS5 (AMETEK) universal testing machine, a single column benchtop system fitted with a 5 kN load cell and mechanical wedge-action grips with serrated jaw faces. The main components of the testing system are shown in Figure 3.4. The machine was operated using NEXYGEN Plus 4.0 software for test control and data acquisition. Strain during the modulus-determination stage was measured using an Epsilon Technology Corp clip-on axial extensometer (Model 3542-025M-010-ST). The extensometer was mounted directly across the specimen gauge section and was removed before the pull-to-break stage. Strain during the pull to break stage was therefore determined from crosshead displacement.



Figure 3.4 Lloyd LS5 tensile testing machine with Epsilon extensometer.

3.4.2 Testing Procedure

Testing was conducted in accordance with BS EN ISO 527-2:2025, using NEXYGEN Plus software for test control and data acquisition. A nominal specimen cross-sectional area of 2 mm × 4 mm was used for stress calculations throughout. The validity of this nominal area is examined further in Section 3.5.3.

Testing was carried out in two sequential stages on each specimen using separate test configurations. The first stage was used to determine Young's modulus. An extensometer was clipped onto the specimen, and a pre-load of 5 N was applied at 1 mm/min. A specified gauge length of 25 mm was used for strain measurement. The crosshead was then driven at an extension rate of 1 mm/min, and the stage was stopped at a strain of 0.5%.

The extensometer was then removed, and the same specimen was continued to failure using the second test configuration. A pre-load of 0.5 N was applied at 5 mm/min, and strain was calculated from crosshead displacement using a specified gauge length of 45.41 mm. The crosshead was then driven at an extension rate of 5 mm/min until specimen failure. The resulting data were used to determine ultimate tensile strength, elongation at break and toughness. This two-stage procedure was applied consistently to all specimens tested.

3.4.3 Data Processing

Stress was calculated for each specimen using the actual measured cross-sectional area. Specimen width and thickness were measured at three points along the gauge section using a micrometer, and the mean width and thickness measurements were used to determine the cross-sectional area:

$$A = \bar{w} \bar{t}$$

Engineering stress was then calculated from the applied tensile load and the measured cross-sectional area:

$$\sigma = \frac{F}{A}$$

This approach replaced the initial calculation based on the nominal design cross-section of 2 mm × 4 mm = 8 mm², allowing variations in the actual specimen dimensions to be accounted for in the stress calculation. Strain was unaffected by this correction, as it was derived from extension and gauge length independently of cross-sectional area.

Young's modulus was used as a measure of the material's stiffness, representing the resistance of the specimen to elastic deformation under tensile loading. It was determined from the first testing

stage using the extensometer-based strain measurement recorded over the initial 0.5% strain range. Young's modulus was calculated from the gradient of the linear stress-strain response within the ISO 527 fitting window of 0.05–0.25% strain:

$$E = \frac{\Delta \sigma}{\Delta \varepsilon}$$

A specimen was excluded from the Young's modulus calculation if its stress-strain data did not fully cover the ISO 527 fitting window of 0.05–0.25% strain, or if the linear-regression coefficient of determination (R^2) within this window was below 0.95. This criterion was applied consistently across both build orientations.

Ultimate tensile strength (UTS) was used as a measure of the maximum tensile stress sustained by a specimen during the pull-to-break test. It was determined as the maximum engineering stress recorded during the test:

$$UTS = \max(\sigma)$$

Elongation at break was used as a measure of the deformation sustained by the specimen before fracture and therefore provided an indication of its tensile ductility. It was determined from the strain recorded at the point of specimen failure:

$$\varepsilon_b = \frac{L_f - L_0}{L_0} \times 100$$

Toughness represents the energy absorbed by the material per unit volume during tensile deformation up to failure. It was calculated as the area under the individual engineering stress-strain curve up to failure:

$$U_T = \int_0^{\varepsilon_b} \sigma(\varepsilon) d\varepsilon$$

For the discrete experimental stress-strain data, the integral was evaluated using the trapezoidal rule:

$$U_T \approx \sum_{i=1}^{n-1} \frac{\sigma_i + \sigma_{i+1}}{2} (\varepsilon_{i+1} - \varepsilon_i)$$

For descriptive classification of failure behaviour, specimens were classified as exhibiting limited/brittle failure or extensive drawing based on elongation at break, using a threshold derived empirically from the observed data. A two-cluster k-means analysis was applied to log-transformed elongation-at-break values across all 40 PETG pull-to-break specimens, identifying two populations:

a lower cluster ($n=28$, 3.92–28.2% elongation) and an upper cluster ($n=12$, 42.8–168% elongation). A threshold of 35.5%, the midpoint between the two clusters, was adopted to distinguish limited/brittle failure ($\leq 35.5\%$ elongation) from extensive drawing ($> 35.5\%$ elongation). Stress-strain curves were subsequently inspected to confirm that specimens classified as exhibiting extensive drawing showed the corresponding extended post-yield drawing behaviour.

Representative specimens were selected for graphical presentation of stress-strain behaviour for each material–K-value–orientation condition, using a composite score calculated from the standardised distance of each specimen's UTS and elongation at break from the corresponding group medians, with a zero-IQR guard applied where necessary. This procedure was applied uniformly across all five specimens within each condition, without filtering by failure mode. The selected curves were subsequently checked visually against the remaining specimens within the same condition to confirm that they were broadly representative of the group.

Statistical analysis was performed in RStudio using R. Young's modulus and UTS were analysed using a three-way analysis of variance (ANOVA) with material, K-value and build orientation as the factors. Elongation at break and toughness were log-transformed prior to ANOVA because of the strongly right-skewed distributions observed in the raw data. The same three-way ANOVA model was applied to the transformed data. Where significant main effects or interactions were identified, Tukey's honestly significant difference (HSD) post-hoc comparisons were used to identify specific pairwise differences. Statistical significance was assessed at ($p < 0.05$).

3.5 Optical Microscopy

3.5.1 Equipment

Optical microscopy was performed using a Nikon SMZ800 binocular stereo zoom microscope, fitted with a Plan 1x objective and a transmitted-light illumination stage. Illumination was provided by an Olympus Highlight 2001 fibre optic light source. Images were captured using a Nikon Digital Sight DS-Fi1 camera, mounted on the microscope's trinocular port and connected via a Nikon Digital Sight control unit to a PC running NIS-Elements F software.

3.5.2 Imaging Procedure and Processing

Pre-test surface and post-test fracture images were captured for all specimens across the 16 experimental conditions. Imaging was performed at both the gauge and grip sections, with the gauge section prioritised as the primary region of interest because it represents the main region

subjected to deformation during tensile testing. Grip-section images were also captured as supplementary evidence of the overall print quality. Representative examples of the captured pre-test surface morphology at selected flow values are shown in Figure 3.5.

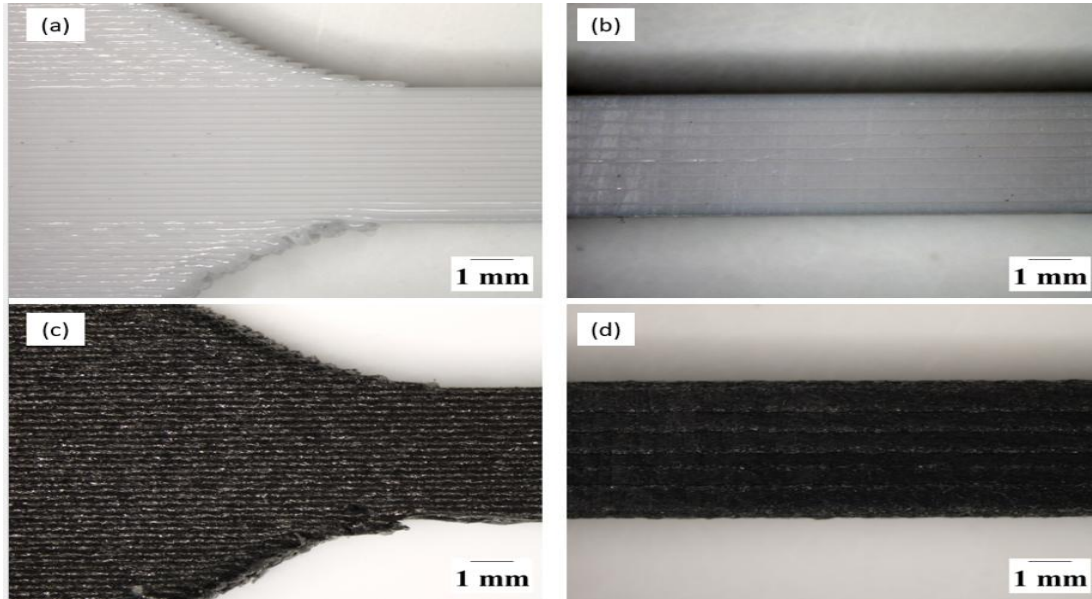


Figure 3.5 Representative pre-test surface morphology of the tensile specimens obtained by Optical imaging at selected flow values: (a) PETG, 100% flow, grip section; (b) PETG, 80% flow, gauge section; (c) CF-PETG, 110% flow, grip section; and (d) CF-PETG, 110% flow, gauge section

Captured images were subsequently processed using ImageJ, where the images were calibrated and scale bars were added. Brightness and contrast were adjusted where necessary to improve image clarity while preserving the underlying specimen features.

4.0 Results

4.1 Physical Characterisation

Physical characterisation was carried out to assess the dimensional and mass changes associated with the different K-values and build orientations prior to evaluating the tensile results. Specimen dimensions were measured after printing to determine the actual cross-sectional area of the gauge section, while specimen mass was recorded to assess changes in material deposition. Optical microscopy was also used to provide qualitative observations of the printed surface morphology.

4.1.1 Dimensional Accuracy

The nominal cross-sectional area of the specimen gauge section was 8.0 mm^2 , based on the nominal dimensions of 4.0 mm width and 2.0 mm thickness. The actual cross-sectional area was determined

from the mean of three width and three thickness measurements taken along the gauge section of each specimen. The measured dimensions showed variation with material, K-value and build orientation.

Overall, PETG specimens remained relatively close to the nominal 8.0 mm² cross-sectional area across the investigated K-values. The deviation from the nominal area ranged from –3.2% to +2.7% across the PETG conditions. In comparison, CF-PETG exhibited larger positive deviations, particularly in the on-edge orientation. The deviation for CF-PETG increased from +4.2% at K = 80% to +13.9% at K = 110% in the on-edge orientation. The corresponding flat-orientation deviations ranged from +2.4% to +6.9%.

The largest measured deviation from the nominal cross-sectional area was therefore observed for CF-PETG in the on-edge orientation at K = 110%, where the measured area was approximately 13.9% greater than the nominal 8.0 mm². This corresponds to an actual mean area of approximately 9.11 mm². The measured cross-sectional area for each material and orientation across the four K-values is shown in Figure 4.1.

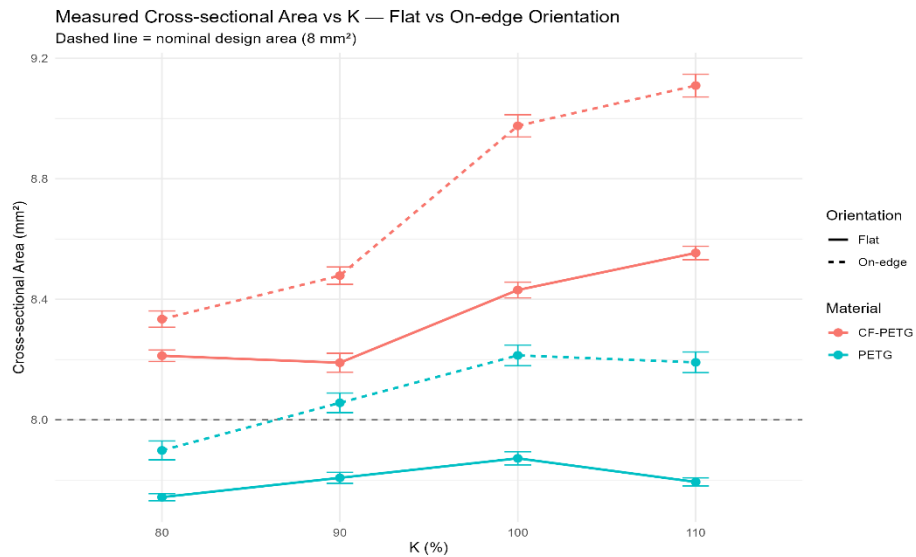


Figure 4.1 Mean measured cross-sectional area vs K-value. Error bars represent $\pm$ SD; dashed line indicates the 8.0 mm² nominal area.

4.1.2 Mass Variation with K

The measured specimen mass varied with material, build orientation and K-value. For each experimental condition, the mean mass and standard deviation were calculated from the five tested specimens. PETG generally exhibited higher specimen masses than CF-PETG in the flat orientation, while the difference between the materials was smaller in the on-edge orientation.

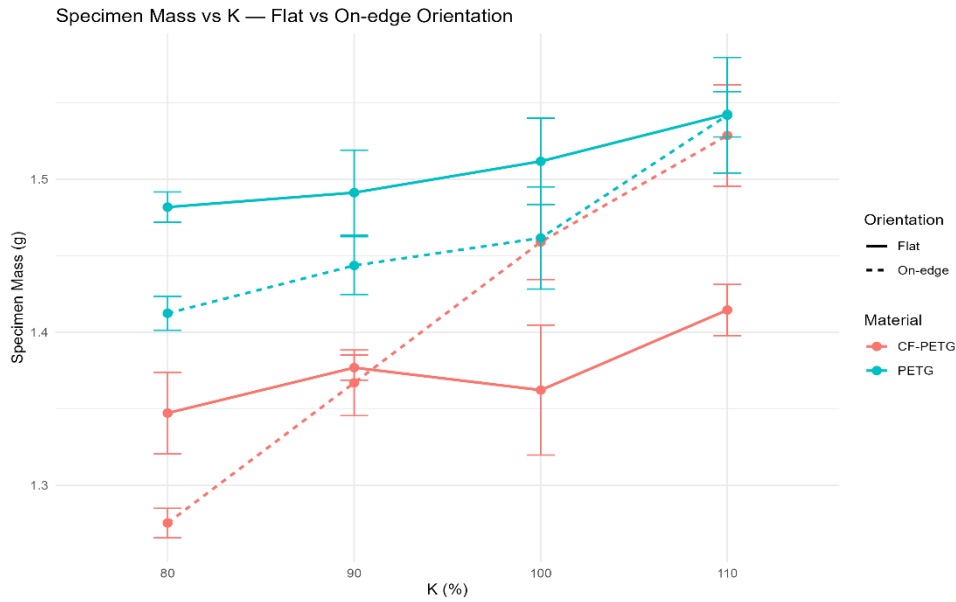


Figure 4.2 Mean specimen mass vs K-value. Error bars represent $\pm$ SD.

For PETG in the flat orientation, the mean mass increased from 1.482 g at K = 80% to 1.542 g at K = 110%. In the on-edge orientation, the mean mass increased from 1.412 g to 1.542 g over the same K range. CF-PETG showed a different trend in the flat orientation, with mean mass values of 1.347, 1.377, 1.362 and 1.415 g at K = 80, 90, 100 and 110%, respectively. In the on-edge orientation, CF-PETG showed a more pronounced increase, from 1.275 g at K = 80% to 1.529 g at K = 110%. The mean specimen mass and associated variation across the four K-values and two build orientations are presented in Figure 4.2.

4.1.3 Optical Microscopy

Representative post-test fracture surfaces were examined for each material and build orientation. The microscopy images were used to document the visible surface and fracture morphology following tensile testing. Representative images for PETG and CF-PETG are shown in Figures 4.3 and 4.4, respectively.

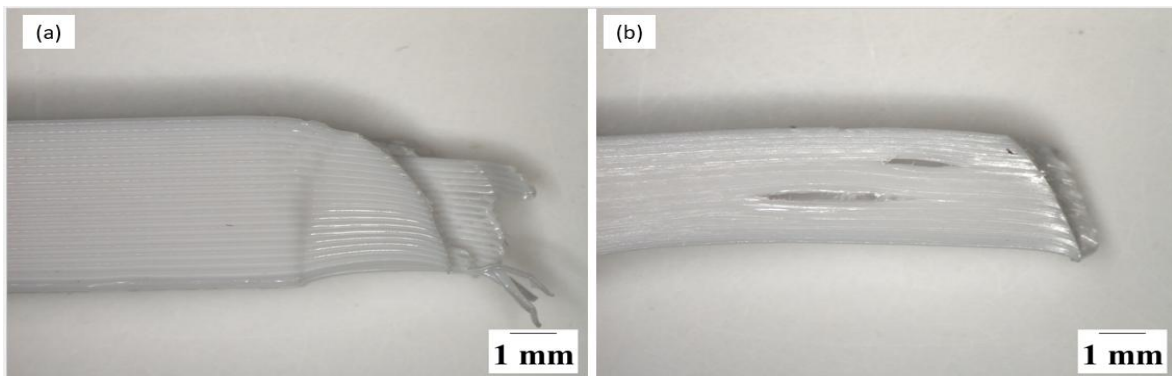


Figure 4.3 Optical micrographs of post-test PETG fracture surfaces: (a) Flat orientation at K = 100% and (b) On-edge orientation at K = 100%.

The PETG fracture surfaces showed visible printed layer and raster features, with variations in the appearance of the fracture edge between specimens. Some surfaces exhibited relatively stepped fracture features, while others showed localised deformation and extended polymer strands at the fracture edge. The representative PETG image at $K = 100\%$ also contained two elongated dark features within the specimen surface; however, their origin could not be confirmed from optical microscopy alone and they were therefore not classified as porosity.

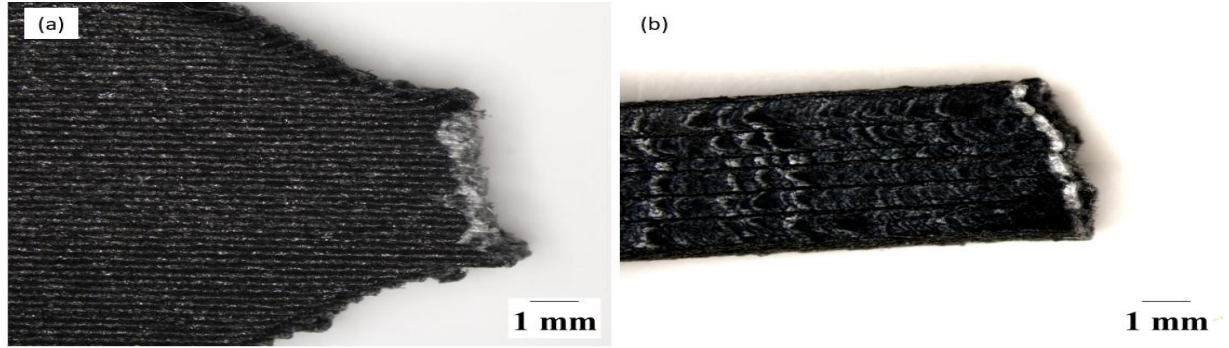


Figure 4.4 Optical micrographs of post-test CF-PETG fracture surfaces: (a) Flat orientation at $K = 110\%$ and (b) On-edge orientation at $K = 100\%$.

The CF-PETG fracture surfaces similarly retained visible printed-layer features, with fibrous protrusions and irregular material features observed near the fracture edge. The fracture surfaces also showed pronounced banding corresponding to the printed layers. No obvious large-scale interlayer separation was observed in the representative images examined. The optical images provide qualitative evidence of the fracture morphology, while detailed assessment of fibre–matrix bonding or individual fibre failure would require higher-resolution techniques. Specimen

4.2 Tensile Test Results

This section presents the tensile properties determined from the two-stage testing procedure: Young's modulus, ultimate tensile strength, elongation at break, and toughness, followed by the failure-mode classification. Differences between materials, K -values, and orientations were assessed using a three-way ANOVA (Material $\times$ $K \times$ Orientation) for each property, with Tukey post-hoc comparisons used to identify specific pairwise differences where a significant interaction was found, as described in Section 3.5.3.

4.2.1 Representative Stress–Strain Curve

For each material– K -value–orientation condition, a single representative specimen was selected to illustrate characteristic stress–strain behaviour, using the composite-score method described in Section 3.4.3. Representative curves for CF-PETG in the flat and on-edge orientations are shown in Figures 4.5 and 4.6, respectively, while the corresponding PETG curves are shown in Figures 4.7 and 4.8.

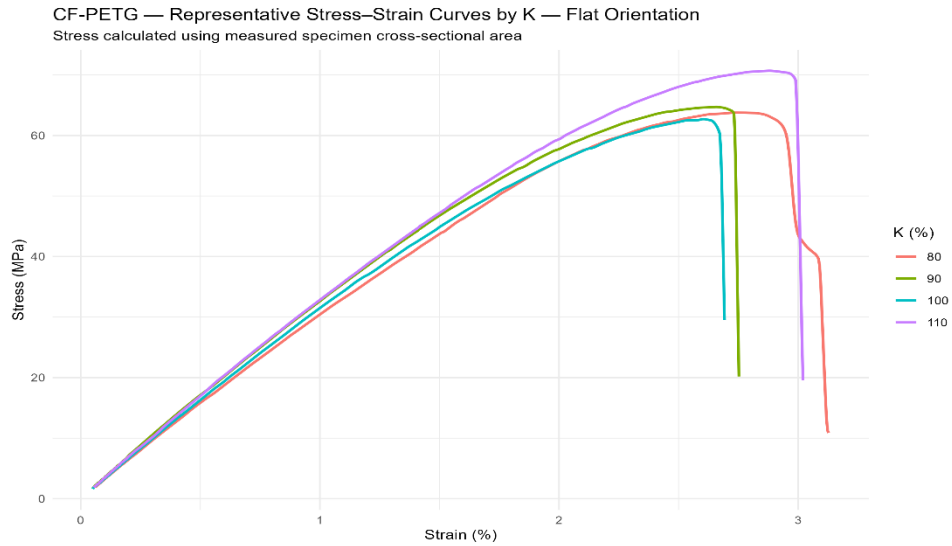


Figure 4.5 Representative stress–strain curves for CF-PETG in the flat orientation at K = 80–110%.

The representative CF-PETG curves show broadly similar stress–strain behaviour across the four K conditions within each orientation. In the flat orientation (Figure 4.5), the curves reach peak stresses of approximately 62–70 MPa, followed by a relatively rapid reduction in stress towards failure. The K = 110% curve reaches the highest peak stress and extends to the greatest strain among the four representative curves. In the on-edge orientation (Figure 4.6), the curves reach peak stresses of approximately 60–74 MPa and show a relatively abrupt reduction in stress following the peak, with failure occurring at approximately 2.5–3.3% strain.

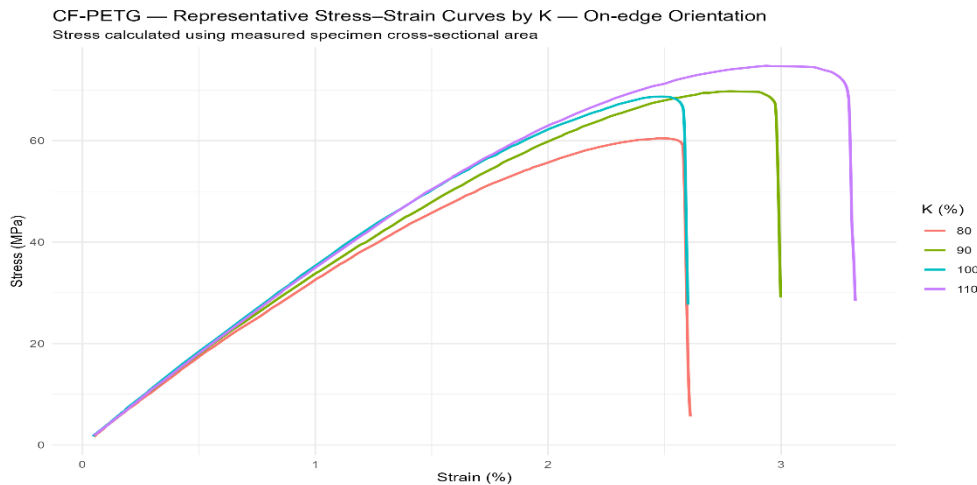


Figure 4.6 Representative stress–strain curves for CF-PETG in the On-edge orientation at K = 80–110%

The PETG representative curves show greater variation in post-peak behaviour across the tested conditions. In the flat orientation (Figure 4.7), the K = 80% and K = 90% curves show a pronounced reduction in stress following the peak and subsequently terminate at relatively low strains. The K = 100% curve exhibits a substantial initial stress reduction followed by an extended, relatively stable

post-peak region that continues beyond the plotted strain range. In contrast, the K = 110% curve shows a moderate post-peak reduction followed by a gradual decline in stress until final failure. The extended strain range used for the flat-orientation PETG plot allows this longer post-peak response to be visualised.

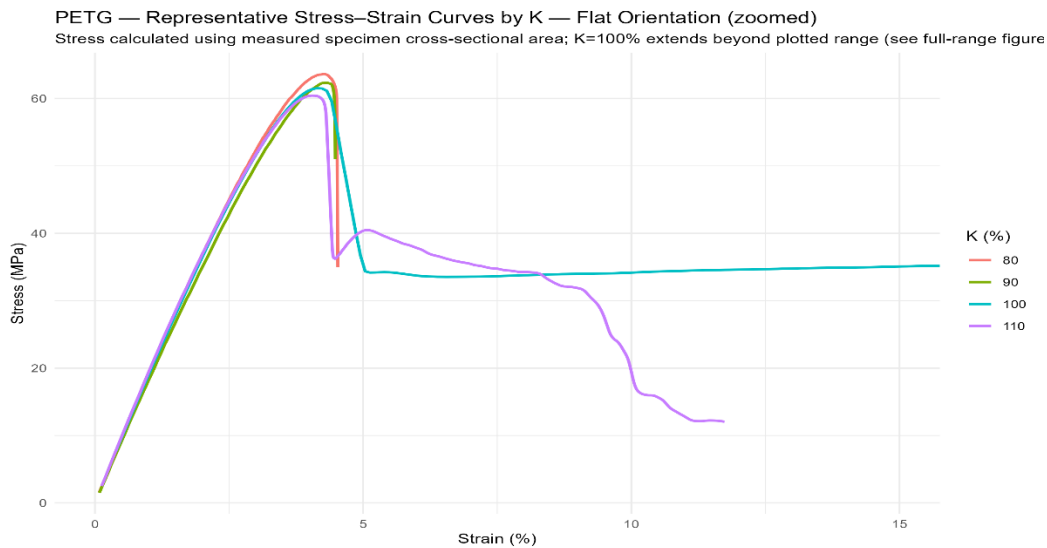


Figure 4.7 Representative stress–strain curves for PETG in the flat orientation at K = 80–110%
 For PETG in the on-edge orientation (Figure 4.8), all four representative curves show an extended post-peak deformation region following the initial peak stress. The curves subsequently undergo a gradual reduction in stress over a substantially greater strain range than the CF-PETG curves, with the representative curves extending to approximately 16% strain. The K = 110% curve extends slightly further than the other representative curves.

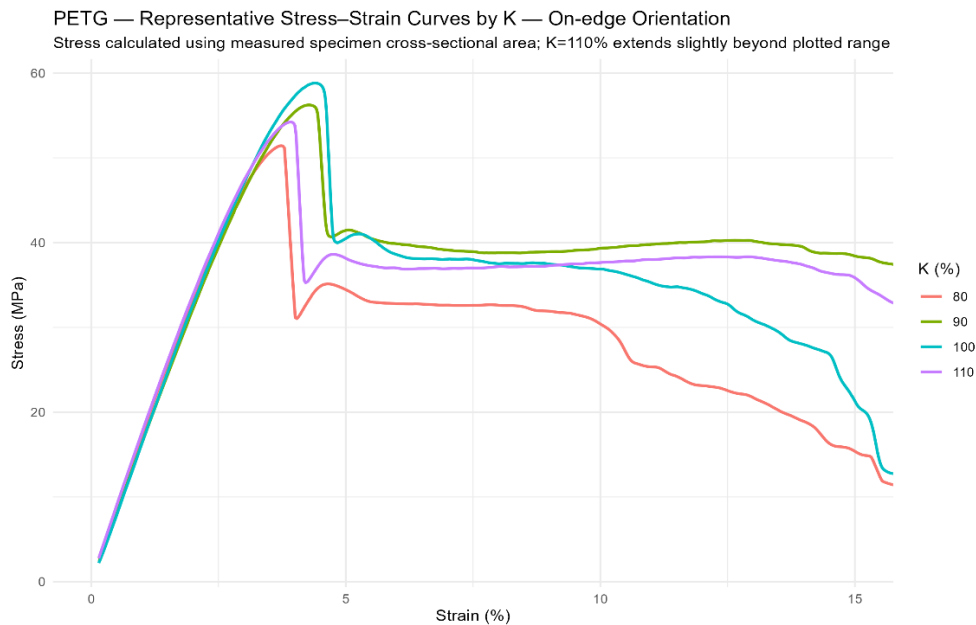


Figure 4.8 Representative stress–strain curves for PETG in the On-edge orientation at K = 80–110%

The PETG specimens exhibited both limited/brittle failure and extensive drawing. Specimens were classified based on a threshold of 35.5% elongation at break, as described in Section 3.4.3. CF-PETG specimens were classified as limited/brittle in 19 of 20 tests across both orientations combined, with a single exception (flat orientation, K = 80%, Sample 5). PETG showed a more varied failure-mode distribution, summarised in Table 4.1.

Table 4.1 Number of PETG specimens classified as exhibiting extensive drawing by K-value and build orientation

K-Value	Flat Orientation	On-Edge
80%	0/5	1/5
90%	2/5	1/5
100%	3/5	0/5
110%	2/5	3/5

Several conditions showed a near-even (3:2) division between the two failure modes — flat K = 90%, flat K = 100%, flat K = 110%, and on-edge K = 110% indicating that neither failure behaviour was clearly dominant at these conditions.

Overall, the representative curves illustrate differences in the shape and extent of the tensile response between the two materials and orientations. Quantitative comparisons of Young's modulus, ultimate tensile strength, elongation at break and toughness are presented separately in Sections 4.2.2–4.2.5.

4.2.2 Young's Modulus

Young's modulus was determined from the initial linear region of the stress–strain response using the extensometer-based strain measurement. The mean Young's modulus values for each material, K-value and build orientation are presented in Figure 4.9, with error bars indicating the variability within each condition.

For CF-PETG, Young's modulus was consistently higher in the on-edge orientation than in the flat orientation across all K-values (Figure 4.9). In the flat orientation, the mean Young's modulus ranged from 4.64 to 4.97 GPa, with the highest value at K = 90%. The modulus increased from 4.77 GPa at K = 80% to 4.97 GPa at K = 90%, remained similar at 4.92 GPa for K = 100%, and decreased to 4.64 GPa at K = 110%. In the on-edge orientation, the modulus ranged from 5.59 to 6.00 GPa. The highest value occurred at K = 90% (6.00 GPa), while values of 5.59, 5.61 and 5.75 GPa were obtained at K = 80%, 100% and 110%, respectively.

PETG exhibited substantially lower Young's modulus values than CF-PETG in both orientations. For the flat orientation, the mean modulus increased gradually from approximately 2.14 GPa at K = 80% to approximately 2.29 GPa at K = 110%, with only a small change between K = 80% and K = 90%. In the on-edge orientation, the modulus remained close to 2.0 GPa from K = 80% to K = 100%, with values of approximately 1.96, 1.97 and 1.98 GPa, respectively, before increasing to approximately 2.09 GPa at K = 110%.

Statistical analysis showed a significant material × orientation interaction for Young's modulus ($F = 90.90$, $p < 0.001$). Tukey post-hoc comparisons showed that the Young's modulus of CF-PETG was significantly higher in the on-edge orientation than in the flat orientation, with a mean difference of 0.91 GPa ($p < 0.001$). In contrast, the difference between the flat and on-edge orientations for PETG was not statistically significant ($p = 0.086$).

Overall, CF-PETG showed substantially greater stiffness than PETG across all K-values and orientations. The statistically significant material × orientation interaction indicates that the effect of build orientation on Young's modulus differed between the two materials. Variations in K produced comparatively modest changes in Young's modulus within each material.

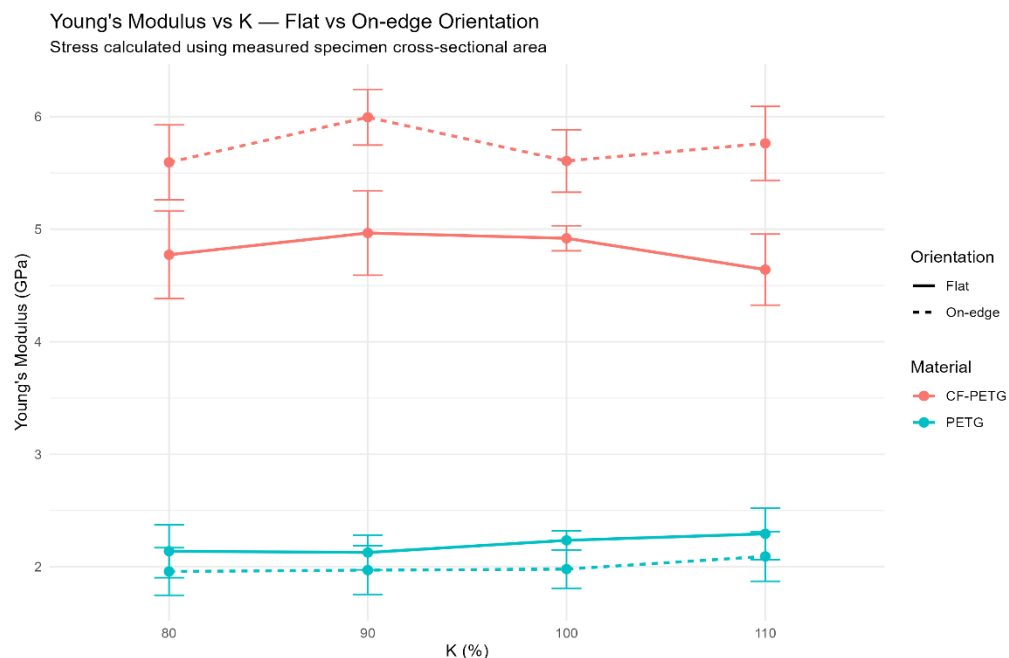


Figure 4.9 Mean Young's Modulus vs K— Flat vs On-edge

4.2.3 Ultimate Tensile Strength (UTS)

Ultimate tensile strength (UTS) was determined as the maximum stress recorded during each tensile test. The mean UTS values for each material, K-value and build orientation are presented in Figure 4.10, with error bars representing the variability within each condition.

For CF-PETG, UTS showed greater variation with K-value than was observed for Young's modulus. In the flat orientation, the mean UTS was approximately 65 MPa at K = 80%, decreased slightly to approximately 64 MPa at K = 90%, and reached a minimum of approximately 62 MPa at K = 100%. At K = 110%, the mean UTS increased substantially to approximately 69 MPa. In the on-edge orientation, UTS increased from approximately 60 MPa at K = 80% to approximately 67 MPa at K = 90% and approximately 68 MPa at K = 100%, before reaching approximately 75 MPa at K = 110%.

PETG showed comparatively smaller changes in UTS across K-values, particularly in the flat orientation. The mean UTS for flat specimens remained close to 63 MPa across the four K-values, with values of approximately 63 MPa at K = 80%, 63 MPa at K = 90%, 62 MPa at K = 100% and 63 MPa at K = 110%. In the on-edge orientation, the mean UTS was lower, remaining approximately 54 MPa at K = 80% and K = 90%, increasing slightly to approximately 55 MPa at K = 100%, and reaching approximately 57 MPa at K = 110%.

Statistical analysis showed significant main effects of material ($F = 68.54$, $p < 0.001$) and K ($F = 5.88$, $p = 0.001$) on UTS. The material $\times$ K interaction was also significant ($F = 3.69$, $p = 0.016$), as was the material $\times$ orientation interaction ($F = 33.18$, $p < 0.001$). The K $\times$ orientation interaction was not statistically significant at the conventional threshold, although it was borderline ($F = 2.69$, $p = 0.054$). The three-way material $\times$ K $\times$ orientation interaction was not significant ($F = 1.32$, $p = 0.276$). Tukey post-hoc comparison of the material $\times$ orientation interaction showed that PETG had significantly lower UTS in the on-edge orientation than in the flat orientation, with the on-edge value 8.35 MPa lower ($p < 0.001$). For CF-PETG, the on-edge orientation showed a 2.18 MPa higher UTS than the flat orientation, but this difference was not statistically significant ($p = 0.338$).

Tukey comparison of the material $\times$ K interaction, pooled across orientation, showed that CF-PETG UTS at K = 110% was significantly higher than at K = 80%, K = 90% and K = 100% (all $p < 0.001$); no other pairwise K comparisons for CF-PETG reached significance. None of the six pairwise K comparisons for PETG were statistically significant (all $p > 0.94$). Descriptively, the effect of K on CF-PETG UTS held broadly across both orientations, though the K $\times$ orientation interaction ($p = 0.054$) suggests that the magnitude of this effect may have differed somewhat between the two orientations.

Overall, CF-PETG and PETG exhibited different responses to K-value and build orientation. CF-PETG reached its highest UTS at K = 110%, with the increase from the lower K-values being statistically significant in the pooled material $\times$ K comparison. PETG showed comparatively little variation in UTS

across K-values, while its UTS was significantly lower in the on-edge orientation than in the flat orientation.

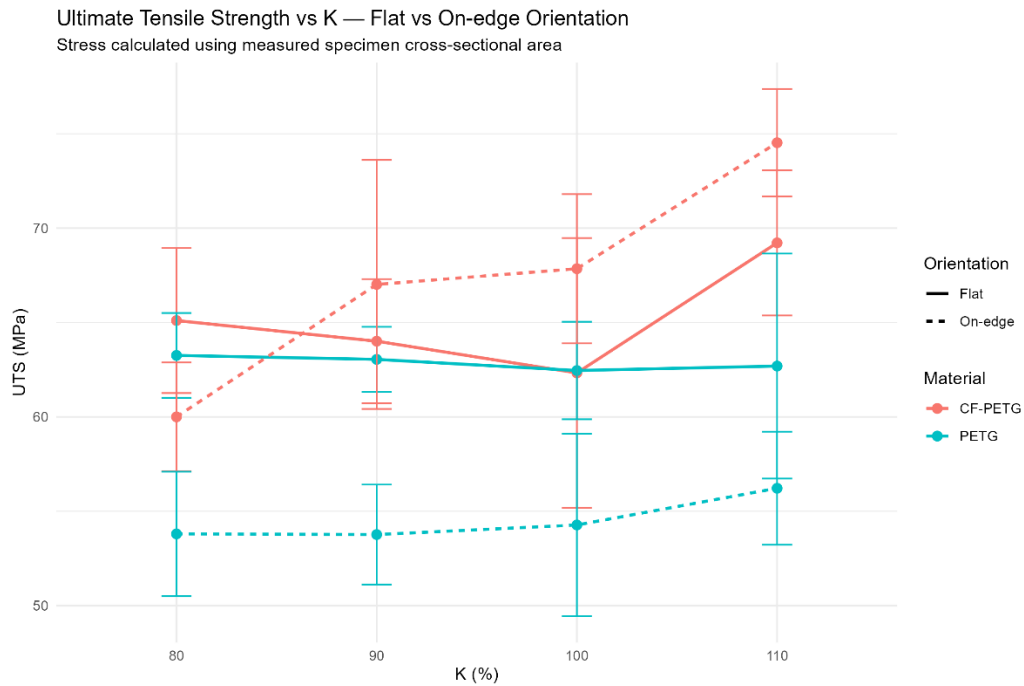


Figure 4.10 Mean UTS vs K— Flat vs On-edge

4.2.4 Elongation at Break

Elongation at break showed substantially greater variability for PETG than for CF-PETG, as shown in Figure 4.11. CF-PETG exhibited consistently low elongation across both orientations and all K-values, with median values remaining close to 3%.

PETG showed markedly greater elongation and stronger variation with K and orientation. In the flat orientation, median elongation remained relatively low at K = 80% and 90%, increased substantially at K = 100%, and decreased again at K = 110%. The on-edge orientation showed a different pattern, with median elongation remaining in the tens of percent across all K-values and increasing notably at K = 110%. The large IQRs for several PETG conditions indicate substantial specimen-to-specimen variability. The tested specimens were classified as exhibiting either limited/brittle failure or extensive drawing using the 35.5% elongation-at-break threshold. The resulting distribution of extensive-drawing specimens is given in Table 4.5. CF-PETG was predominantly classified as limited/brittle, whereas PETG contained both behaviours, with the proportions varying between K-values and orientations.

Statistical analysis of log-transformed elongation values showed a significant main effect of material ($F = 125.81$, $p < 0.001$) and a significant material $\times$ K $\times$ orientation interaction ($F = 4.33$, $p = 0.008$), indicating that the effect of K on elongation depended jointly on material and orientation. Neither the orientation main effect nor the material $\times$ orientation interaction was statistically significant (both $p > 0.05$). Overall, elongation at break distinguished the two materials more clearly than the other tensile properties, with CF-PETG showing consistently limited elongation and PETG exhibiting substantially greater and more variable deformation.

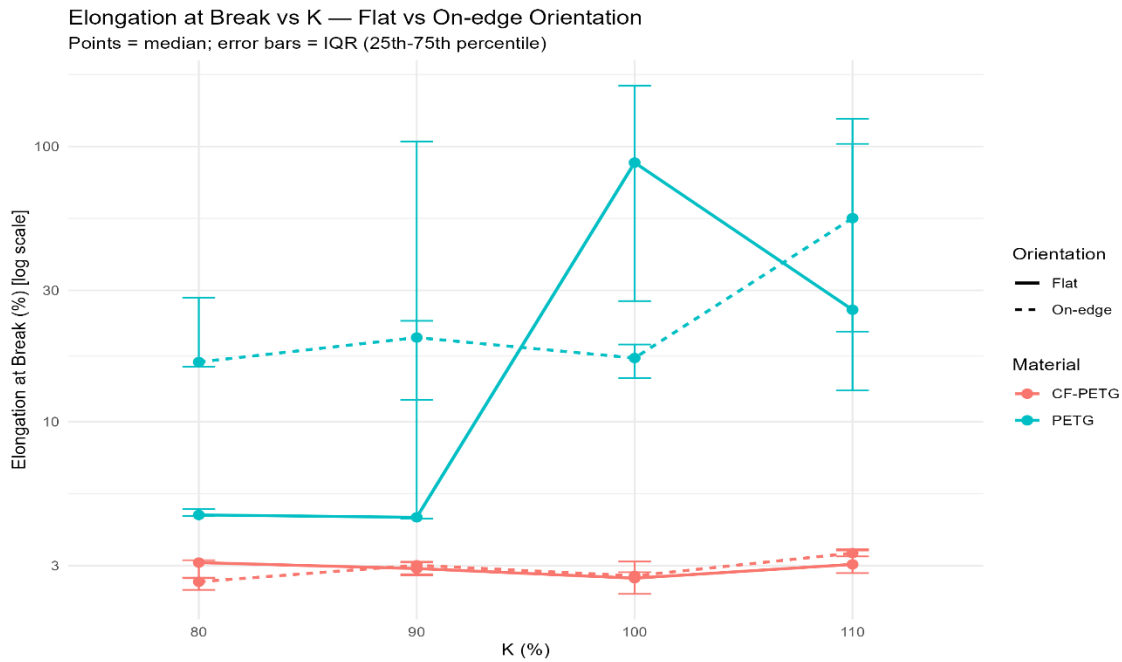


Figure 4.11 Elongation at Break vs K— Flat vs On-edge

4.2.5 Toughness

Toughness showed substantially greater variability for PETG than for CF-PETG, with median values and IQRs plotted in Figure 4.12. CF-PETG exhibited relatively low toughness across both orientations, with median values remaining close to 1–1.5 MJ/m³ across the K range. PETG showed considerably higher toughness, particularly in the flat orientation at K = 100%, where the median exceeded 30 MJ/m³. In the on-edge orientation, PETG toughness increased to its highest median at K = 110%, reaching approximately 20 MJ/m³. The large IQRs observed for PETG at several conditions reflect the substantial specimen-to-specimen variability associated with the mixed failure-mode composition described in Section 4.2.1.

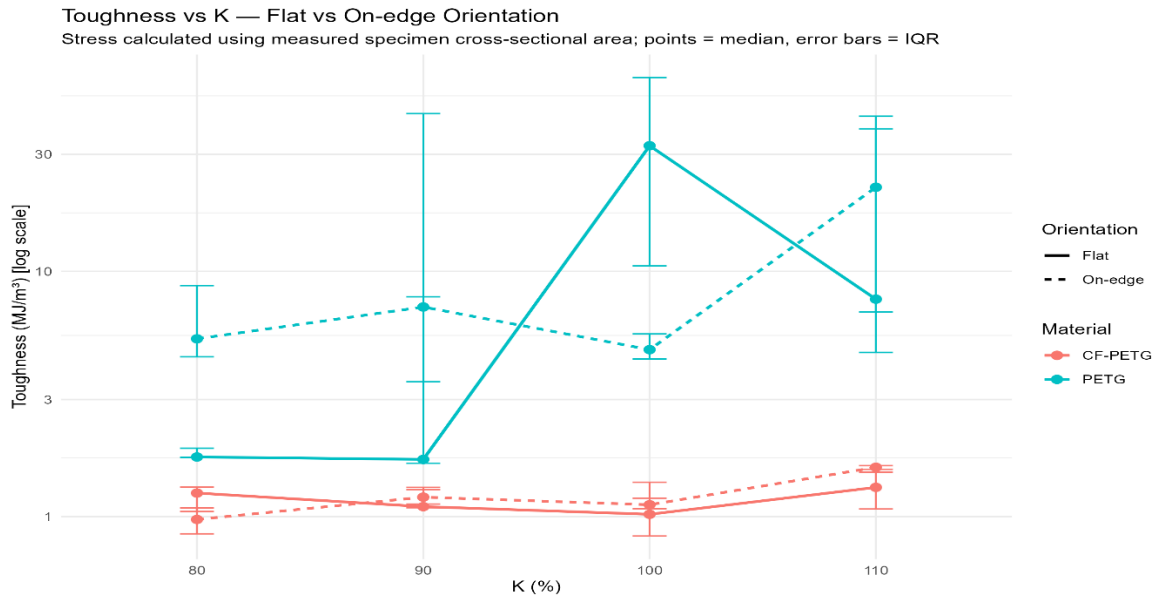


Figure 4.12 Toughness vs K— Flat vs On-edge

Statistical analysis of log-transformed toughness values showed a significant main effect of material ($F = 104.77$, $p < 0.001$) and K ($F = 4.28$, $p = 0.008$). A significant material $\times$ K interaction was also observed ($F = 3.14$, $p = 0.031$), while the material $\times$ K $\times$ orientation interaction was significant ($F = 4.35$, $p = 0.008$), indicating that the combined influence of K and orientation differed between materials. Neither the main effect of orientation ($F = 0.03$, $p = 0.858$) nor the material $\times$ orientation interaction ($F = 0.08$, $p = 0.776$) was statistically significant. The K $\times$ orientation interaction was also not significant ($F = 2.28$, $p = 0.088$). Overall, toughness was strongly dependent on material, with PETG exhibiting substantially greater and more variable toughness than CF-PETG. The significant three-way interaction indicates that the pattern of toughness across K and orientation was material dependent.

5.0 Discussion

The results demonstrate that the effect of flow-rate multiplier (K) on mechanical performance was material-dependent. CF-PETG showed a clearer response to changes in K, particularly in ultimate tensile strength, while PETG exhibited comparatively stable strength and stiffness but substantially greater variation in elongation and toughness. The results also indicate that the influence of K cannot be considered independently of material and, for some properties, build orientation.

5.1 Effect of Flow Rate on Mechanical Performance

The effect of K on mechanical performance was material dependent. CF-PETG showed the clearest response to increasing flow rate, with $K = 110\%$ producing significantly higher UTS than $K = 80\%$, 90% and 100% . In contrast, Young's modulus showed relatively little variation with K , suggesting that flow rate had a greater influence on ultimate strength than the initial elastic response.

The increase in CF-PETG UTS at $K = 110\%$ may be associated with improved deposition continuity and reduced gaps between adjacent extrusion paths. Increasing the material flow can provide greater filling of the intended extrusion path, potentially improving contact between neighbouring deposited regions and reducing local stress-concentrating discontinuities. This is consistent with Minaoglou et al. (2024), who similarly reported that flow rate contributed to changes in CF-PETG tensile performance, although their study varied flow rate alongside other printing parameters. Microscopy revealed local irregularities and discontinuities between deposited paths (Figure 5.1), providing qualitative evidence of defects within the printed structure. However, the image alone cannot establish whether their occurrence or severity changed systematically with K .

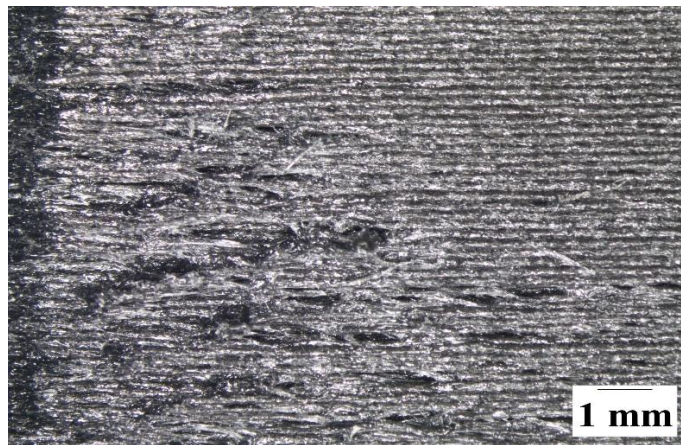


Figure 5.1 CF-PETG surface showing local deposition irregularities

PETG showed comparatively stable UTS and modulus but substantially greater variation in elongation and toughness. The representative stress–strain curves showed that PETG specimens could undergo either limited post-peak deformation or extensive drawing, depending on the tested condition. This difference in deformation behaviour helps explain the larger spread observed in elongation and toughness compared with CF-PETG. The greater variability also indicates that increasing flow rate does not necessarily produce a uniform improvement across all mechanical properties.

Build orientation produced some visible differences, particularly in the CF-PETG UTS response, where the on-edge specimens showed a stronger descriptive increase towards $K = 110\%$. However, the $K \times$ orientation interaction was not statistically significant ($p = 0.054$), so this should be regarded as a trend rather than a confirmed orientation-specific effect.

Overall, $K = 110\%$ provided the strongest UTS response for CF-PETG within the investigated range, while PETG showed a less pronounced response in strength and stiffness but greater variability in deformation and toughness. Therefore, K-value selection should consider both the material and the mechanical property required rather than assuming that increased flow universally improves performance.

5.2 Sustainability

K-Value optimisation has the potential to improve the resource efficiency of FDM manufacturing. Selecting an appropriate K-value can reduce unnecessary over- or under-extrusion and the likelihood of poorly formed or failed components. This can reduce material waste and the additional energy associated with producing replacement parts. Optimising the process also provides a more systematic alternative to repeated trial-and-error printing when establishing suitable manufacturing conditions.

The findings therefore support SDG 9 – Industry, Innovation and Infrastructure, through the optimisation of an additive manufacturing process and the development of evidence-based processing conditions. The work also aligns with SDG 12 – Responsible Consumption and Production, as improved control of material deposition can contribute to more efficient material use and reduced manufacturing waste. There is also an indirect connection to SDG 13 – Climate Action, since reducing wasted material and unnecessary reprinting could reduce the associated energy demand and environmental impact.

The sustainability benefit is particularly relevant to additive manufacturing because the ability to control material deposition is one of the key advantages of the process. However, the present investigation did not directly measure material consumption, printing time, electrical energy or carbon emissions. Therefore, the potential reductions in waste and energy should not be interpreted as quantified environmental improvements. Future work incorporating these measurements would allow the environmental benefits of flow-rate optimisation to be evaluated more accurately.

5.3 Limitations and Future Work

Several limitations should be considered when interpreting the results. The tensile tests were conducted using ISO 527-5A specimens with a relatively simple and standardised geometry. Although this provided controlled conditions for comparing K-values, the behaviour observed may not directly represent components with more complex geometries. Features such as corners, holes, thin sections and changes in cross-section could alter material deposition and introduce local stress concentrations. Consequently, the optimum K-value identified from standard tensile specimens may not necessarily provide the same performance in a practical component.

The study investigated only four K-values (80–110%) and two build orientations, limiting the range over which an optimum processing condition could be identified. A wider and more finely spaced range of K-values would allow the response around the observed high-performing conditions to be examined in greater detail. Testing intermediate values could also determine whether the changes in performance occur gradually or whether particular K-values produce more pronounced changes. In addition, PETG and CF-PETG were printed using separate physical printers. Although this allowed each material to be processed using suitable equipment, machine-to-machine variation may contribute to some of the observed differences between materials. Future work using the same printer, where compatible, or multiple calibrated machines would help separate material effects from equipment-related variation.

Optical microscopy identified local defects and irregularities within the printed structure, including discontinuities between deposited paths (Figure 5.1). However, the microscopy was qualitative and could not establish a direct relationship between defect formation and K-value. It also could not reliably quantify defect size, distribution or volume, or confirm mechanisms such as fibre–matrix debonding. Higher-resolution techniques such as scanning electron microscopy, combined with quantitative defect or porosity analysis, could provide greater insight into how K influences the internal structure and fracture behaviour.

Future work should therefore combine mechanical testing, detailed defect characterisation and process monitoring. Testing more complex geometries would establish whether the trends observed with ISO 527-5A specimens remain applicable to practical components. Direct measurements of material consumption, printing time and energy use would also allow the sustainability implications of K-value optimisation to be quantified. Together, these investigations

could provide a stronger basis for selecting K-values that balance mechanical performance, manufacturing reliability, material efficiency and environmental impact.

6.0 Conclusion

- A detailed literature review was conducted covering FDM additive manufacturing, PETG and CF-PETG materials, K effects, build orientation and the mechanical behaviour of printed components. The review identified flow rate as an important processing parameter that can influence material deposition, interlayer bonding and the resulting mechanical performance of FDM-printed parts.
- The influence of flow-rate multiplier (K) on PETG and CF-PETG was experimentally investigated using four K-values (80%, 90%, 100% and 110%) and two build orientations. Young's modulus, ultimate tensile strength, elongation at break and toughness were evaluated using tensile testing, supported by representative stress–strain curves and optical microscopy.
- It was found that CF-PETG exhibited substantially higher Young's modulus than PETG across the investigated conditions and showed the clearest response to increasing K in terms of UTS. K = 110% produced significantly higher CF-PETG UTS than K = 80%, 90% and 100%, while Young's modulus showed comparatively little variation with K.
- PETG showed a less pronounced response to K in strength and stiffness but considerably greater variation in elongation and toughness. This was associated with the presence of both limited/brittle and extensive-drawing behaviour among the specimens. The significant material $\times$ K $\times$ orientation interactions for elongation and toughness demonstrated that these properties depended on the combination of material, K and orientation.
- Differences between flat and on-edge orientations were observed for some conditions, particularly in the CF-PETG UTS response, although the K $\times$ orientation interaction was not statistically significant ($p = 0.054$). Optical microscopy also identified local irregularities and discontinuities within the printed structure, but these observations were qualitative and could not establish a direct relationship between defects and K.

- The investigation demonstrated that K-value selection should be material- and performance-specific. Within the conditions investigated, $K = 110\%$ provided the strongest UTS response for CF-PETG, while PETG showed greater variability in deformation and toughness. Flow-rate optimisation therefore has potential to improve manufacturing efficiency and reduce unnecessary material waste, although these sustainability benefits were not directly quantified in this study.

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Appendix A- Project Management

Gantt Chart

MEE7012-SUM		K-Value Impact on PETG & CF-PETG Workplan Chart															
Student: Vipul Kohinkar		Module: MEE7012-SUM				Academic Year: 2025-26											
#	Task / Work Package	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12	W13	W14	W15	W16
	Week commencing →	25 May	01 Jun	08 Jun	15 Jun	22 Jun	29 Jun	06 Jul	13 Jul	20 Jul	27 Jul	03 Aug	10 Aug	17 Aug	24 Aug	31 Aug	07 Sep
		Work Task		Milestone ▲		University Closure											
PLANNING & ADMINISTRATION																	
1	Literature Review																
2	Risk Assessment Form																
3	Project Description Form																
▲	Milestone: Risk Assessment Submitted																
MATERIALS & PRINTER SETUP																	
4	Material Order (PETG & CF-PETG)																
5	CAD Design – ISO 527-2 Type 5A Dog Bone																
6	Printer Setup & K Value Calibration																
▲	Milestone: Materials Received																
3D PRINTING OF SPECIMENS																	
7	Print PETG & CF-PETG Samples																
▲	Milestone: All Samples Printed																
TESTING & ANALYSIS																	
8	Tensile Testing																
9	Optical Microscopy																
10	University Closure (10 & 13 July)																
11	Data Analysis & Results Processing																
▲	Milestone: Experimental Work Complete																
REPORT & PRESENTATION																	
12	Report Writing																
13	Report Sub section outline submission																
14	Draft Report to Supervisor and Feedback																
12	Final Report Submission (27 Aug)																
▲	Milestone: Report Submitted																
16	Presentation preparation																
17	Presentation submission																
18	Compulsory Presentation/ Interview																

Project Minutes Review

Project Minutes ID - 1766 - 23/06/2026 10:07

Student - Vipul Kohinkar

Staff Member - Oana Istrate

Meeting Type - General Project Meeting

Current Meeting Location and Date - Ashby 6.020 - 27/05/2026

Review of actions from previous meeting

1. Discussion on project: 3D printing type, material used, mechanical and structural properties 2. Suggested reading different literature reports and articles regarding project.

Discussion, Decisions, Assignments

1. Discussed the baseline of project 2. Discussed the timeline and scope for the 15-week project, including upcoming key deadlines. 3. The Ultimaker S5 3D printer will be used for manufacturing the test samples. 4. Reviewed the requirement of testing 7 samples per configuration, totalling 28 neat PETG and 28 CF-PETG samples (56 total). 5. Define objectives as the project goes on. 6. Confirmed that the manufactured test samples will follow the ISO 527-2 Type 5A dog bone geometry.

Agreed Actions and Target Completion Dates

1. Prepare and submit Risk Assessment form by June 2nd. 2. Research and prepare different K-values for discussion at the next meeting. 3. Complete 3D printing and sample manufacturing by the end of June and keep 2 weeks buffer for it. 4. Conduct a meeting with Troy for 3D printing setup and with John for tensile testing. 5. Prepare a presentation for every meeting showing the project aim, objectives, and an explanation of the testing.

Next Meeting Location, Date and Time - Ashby 6.020 - 11/06/2026@ 11:30

Project Minutes ID - 1794 - 23/06/2026 13:10

Student - Vipul Kohinkar

Staff Member - Oana Istrate

Meeting Type - Project Management Phase 1

Current Meeting Location and Date - Ms Teams - 11/06/2026

Review of actions from previous meeting

1. The risk assessment form was submitted and approved. 2. An initial project timeline was established, and a Gantt chart was created to track key project deadlines. 3. Plans were set to complete the 56 total neat PETG and CF-PETG specimens on the Ultimaker S5 by the end of June. 4. A meeting was held with Troy to discuss the Cura software for the 3D printing setup. 5. The material requisition form was submitted and approved.

Discussion, Decisions, Assignments

1. The project progress presentation was shown, including the project aim, Gantt chart, initial literature review on PETG & CF-PETG, and project challenges & risks. 2. The Gantt chart needs to include the optical microscope phase and extend the literature review section to week 12. 3. The literature review needs to be expanded for the whole project; it was suggested to read research papers on different materials like PLA-CF to find suitable print parameters and confirmed that papers using FFF machines can be considered. 4. Filament requisition is approved for PETG and CF-PETG, but delivery is still pending. 5. The ISO 527-2 Type 5A dog-bone specimen was modeled in SolidWorks and imported into Cura

Agreed Actions and Target Completion Dates

1. Modify Gantt chart to include optical microscopy testing phase and some minor changes. 2. Broaden the literature review to include studies on different 3D print materials to better justify print parameters. 3. Refine and improve project objectives. 4. Check on material delivery and give the filament to Troy for drying. 5. Submit the manufacturing requisition form.

Next Meeting Location, Date and Time - Ashby 6.020 - 22/06/2026@ 15:00

Project Minutes ID - 1805 - 30/06/2026 11:50

Student - Vipul Kohinkar

Staff Member - Oana Istrate

Meeting Type - General Project Meeting

Current Meeting Location and Date - Ashby 6.020 - 22/06/2026

Review of actions from previous meeting

1. Modified the Gantt chart to include the optical microscopy phase and refined the project objectives. 2. Broadened the literature review framework to include a wider range of FFF/FDM materials and parameters. 3. Completed the SolidWorks modeling and Cura slicing for the ISO 527-2 Type 5A specimens.

Discussion, Decisions, Assignments

1. The manufacturing form has been submitted and approved. 2. A printing slot has been booked for this Thursday on the Ultimaker S5 with lab technician Troy. 3. Discussed a few key research papers that gave better insight into the project and helped finalise the printing parameters.

Agreed Actions and Target Completion Dates

1. Email Katie about any forms that need to be filled out or required before doing the tensile testing on the specimens. 2. Start working on the Materials section of the report to prepare for the upcoming submission deadline.

Next Meeting Location, Date and Time - Ashby 6.020 - 03/07/2026@ 11:00

Project Minutes ID - 1815 - 19/07/2026 13:16

Student - Vipul Kohinkar

Staff Member - Oana Istrate

Meeting Type - General Project Meeting

Current Meeting Location and Date - Ashby 6.020 - 03/07/2026

Review of actions from previous meeting

1. Emailed Katie regarding the required documentation for tensile testing and submitted the material testing request form. 2. Started preparing the Materials section of the report in preparation for the project checkpoint submission

Discussion, Decisions, Assignments

1. Reviewed the planned optical microscopy procedure and agreed to examine one specimen from each K-value before and after tensile testing. 2. Discussed the tensile testing outputs to be analysed, including Young's modulus, tensile strength, elongation at break and toughness (area under the stress-strain curve). 3. Discussed the feasibility of printing additional 45° raster angle specimens for comparison and noted that the optimum K-value may differ from the current 0° raster orientation. 4. Discussed the use of ImageJ software for image analysis and the inclusion of a scale bar for all optical micrographs.

Agreed Actions and Target Completion Dates

1. Perform optical microscopy on specimens from each K-value before and after tensile testing. 2. Use ImageJ to analyse optical micrographs and include calibrated scale bars in all images. 3. Carry out tensile testing and evaluate Young's modulus, tensile strength, elongation at break and toughness. 4. Investigate the feasibility of printing specimens with a 45° raster angle for comparison. 5. Prepare and submit the report subsection outline and Materials section for the project checkpoint.

Next Meeting Location, Date and Time - Ms Teams - 22/07/2026@ 10:00

Project Minutes ID - 1835 - 07/08/2026 12:47

Student - Vipul Kohinkar

Staff Member - Oana Istrate

Meeting Type - General Project Meeting

Current Meeting Location and Date - Ms Teams - 22/07/2026

Review of actions from previous meeting

1. Reviewed the optical microscopy procedure for specimens before and after tensile testing. 2. Discussed tensile testing outputs: Young's modulus, tensile strength, elongation at break and toughness. 3. Considered the feasibility of printing additional 45° raster angle specimens for comparison. 4. Discussed the use of ImageJ software and scale bars for optical microscopy analysis.

Discussion, Decisions, Assignments

1. Discussed printing defects observed along the specimen edges and their potential impact on test results. 2. Reviewed the inclusion of calibrated scale bars in optical microscopy images. 3. Discussed the influence of print additives on the mechanical behaviour of the printed materials. 4. Reviewed progress on the project checkpoint report and discussed remaining tasks.

Agreed Actions and Target Completion Dates

1. Add scale bars to all microscopy images. 2. Analyse and discuss printing defects at specimen edges. 3. Complete analysis of tensile testing results. 4. Include a discussion of print additives in the report. 40 Continue writing and finalising the report.

Next Meeting Location, Date and Time - Ashby 6.020 - 07/08/2026@ 15:00

Project Minutes ID - 1860 - 20/08/2026 07:57

Student - Vipul Kohinkar

Staff Member - Oana Istrate

Meeting Type - General Project Meeting

Current Meeting Location and Date - Ashby 6.020 - 07/08/2026

Review of actions from previous meeting

1.Add scale bars to all microscopy images. 2.Analyse and discuss printing defects at specimen edges. 3.Complete analysis of tensile testing results 4.Continue writing and finalising the report.

Discussion, Decisions, Assignments

1.Do not average the data when plotting the stress–strain curves. 2.Use representative stress–strain curves for presentation and comparison. 3.Select the closest representative Young's modulus from the individual specimen results.

Agreed Actions and Target Completion Dates

1.Submit the draft report by 17 August. 2. Complete the RStudio data analysis and generate the required plots.

Next Meeting Location, Date and Time - Ashby 6.020 - 20/08/2026@ 11:30

Project Minutes ID - 1873 - 26/08/2026 11:26

Student - Vipul Kohinkar

Staff Member - Oana Istrate

Meeting Type - General Project Meeting

Current Meeting Location and Date - Ashby 6.020 - 20/08/2026

Review of actions from previous meeting

1.Discussed selection of the most representative Young's modulus from the tensile test results. 2.Agreed that individual stress–strain curves should be presented rather than averaged curves. 3. Submit the draft report by 17Aug

Discussion, Decisions, Assignments

1.Discussed presenting the experimental results using average values and standard deviation. 2.Discussed improving the presentation of results by focusing on the key findings rather than presenting excessive amounts of raw data. 3.Discussed summarising the research gap and linking it back to the reviewed research papers and their findings. 4.Discussed the potential use of Design of Experiments (DOE) in interpreting the project results.

Agreed Actions and Target Completion Dates

1.Calculate and present the average results with standard deviation. 2.Refine the presentation to focus on key findings and avoid unnecessary data. 3.Update the research gap and supporting literature discussion using the reviewed papers. 4.Consider and discuss the suitability of DOE for the project analysis. 5.Prepare a 10-minute presentation and mock interview for the next meeting on 1 September.

Next Meeting Location, Date and Time - Ashby 6.020 - 01/09/2026@ 10:30

Project Submission Checklist

A copy of the checklist can be found on the Student Canvas site and must be included at the very end of your report.

	TICK IF MET
Does the report meet the formatting stipulated in the module handbook and template provided?	☒
Line spacing (1.5).	☒
Font (Calibri 11 Pt).	☒
Margins: Top & Bottom (25 mm), Left (25 mm), Right (25 mm).	☒
Paragraphs are fully justified.	☒
Pages are numbered correctly (initial pages summary, contents etc in roman numerals, Intro – Conclusions, Page 1 – 40).	☒
Does the main body of the report meet the strict 30 page limit (40 page limit for MEE7012)? (excluding Title Page, Table of Contents, Turnitin Summary, References and Submission Checklist)	☒
Do the appendices meet the strict 10 page limit (15 page limit for MEE7012)?	☒
Are all tables and figures numbered correctly, captioned and referenced if required?	☒
Has the report been checked using Turnitin?	☒
Is the Turnitin summary report included in the report?	☒
What is the similarity index provided by Turnitin?	☒

Any use of artificial intelligence (AI) tools, inc. but not limited to ChatGPT, Bing Chat, Google Bard, etc, or related apps (eg QuillBot, etc) is declared in the report.	✓
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Statement of originality

I hereby declare that this project is my own work and that it has not been submitted for another degree, either at Queen's University Belfast or elsewhere. Where other sources of information have been used, they have been acknowledged.

Signature: Vipul Kohinkar

I hereby declare that any use of AI in this project has been documented in the report and related evidence demonstrating how AI was employed to support the project has been included.

Signature: Vipul Kohinkar

Date: 27/08/2026

Documented Use of AI

In the table below please document if you used AI tools during your project. This should be broken into the relevant chapters/sections, what section it was used for and what particular software was used (eg ChatGPT, Bing Chat, Google Bard, etc, or related apps eg QuillBot, etc).

Chapter	Section/Page	What was AI used for in this section?	AI Software Used
1	1.4/ 3	Paraphrasing	Claude,Chatgpt
2	2.5/12	Paraphrasing	Claude
5	5.2/36	Paraphrasing	Chatgpt